

Housing Inequality in Spain*

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October 31, 2025

Abstract

We use data on the universe of real estate in Spain and recent real estate transactions in Catalonia to construct novel contemporaneous and long-run housing inequality estimates for Spain at all existing levels of aggregation. Housing inequality in Spain is between Belgium and the United States, with the top 1% most spacious and valuable dwellings accruing over 6% of total residential space and value. Long-run series indicate that space inequality was much higher at the beginning of the 20th Century and declined significantly in the 1960s, coinciding with a boom in housing supply. Housing value inequality increased between 2009 and 2019, likely due to cyclical changes in aggregate demand and stagnating new housing supply. We document significant heterogeneity across all levels of aggregation and a correlation between housing value and income inequality, suggesting a connection between the two. This research introduces a new publicly available database to study housing-related topics and highlights the importance of granularity when evaluating policies related to inequality and housing.

Keywords: Inequality, Gini, Housing

JEL Codes: D31, D63, O18

*We thank Micael Castanheira and Paula Gobbi for helpful comments and feedback. Gerard Domènech-Arumi gratefully acknowledges funding from Fondation ULB. All errors are our own.

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1 Introduction

Housing is a multifaceted and extremely important good. Housing is a consumption good that provides shelter and access to places (cities and neighborhoods), and it represents the major expenditure category for most households in advanced economies ([OECD 2022a](#)). Housing is also a source of income (rents and imputed rents), and housing is, of course, an asset – the most widespread asset across the majority of households’ portfolios ([Kuhn et al. 2020](#)). Therefore, housing is crucial to understanding the state and dynamics of income, consumption, and wealth inequality ([Dustmann et al. 2022](#), [Martínez-Toledano 2023](#), [Piketty and Zucman 2014](#)). However, despite the importance of housing is widely acknowledged in the literature, we know relatively little about the state or dynamics of inequality in the many facets of housing, across and within countries.

In this paper, we use data on the universe of dwellings in Spain and real estate transaction data from the region of Catalonia to provide new contemporaneous and long-run estimates of housing space (and value, for Catalonia) inequality across all possible levels of aggregation.

The cadastre contains information on all real estate in Spain as of 2023. In the data, we observe the space, construction year, and location of every dwelling. The data does not include information on dwellings’ assessed or market value. Therefore, we rely on transaction data for Catalonia (2009-19) and machine learning techniques to impute a market value to all dwellings in the region over that period. These data allow us to investigate disparities in housing space in Spain and value in Catalonia across all existing levels of aggregation.

Housing space and value inequality in Spain are between those of Belgium and the United States. The top 1% most spacious and valuable dwellings contain 5.8% and 6.5% of all existing residential space and value in Spain and Catalonia, respectively. These translate into a housing space Gini of 0.31 for Spain and a housing value Gini of 0.37 for Catalonia. For comparison, we estimate a space (value) Gini of 0.24 (0.25) in Belgium,

and 0.35 (0.50) in the United States, thus placing Spain between the two countries. These estimates align with those for income, often finding that Spain's inequality levels are in the middle relative to other advanced economies ([Anghel et al. 2018](#)).

Geographic heterogeneity is substantial. The top 1% most spacious dwellings contain 7.1% of total space in Catalonia (Gini of 0.33) but only 3.1% of the space in the Basque Country (Gini of 0.17). Comparably large disparities are observed at lower levels of aggregation, between and within regions. For instance, in terms of value, we document stark disparities between the Barcelona metro area dwellings and those in the rest of the region, with values more than doubling at many points in the distribution, especially above the median. Aggregation matters, as it masks significant heterogeneity.

Housing inequality is not explained by a single factor, but income inequality at low levels of aggregation appears to be tightly connected to it. In the last part of the paper, we investigate the determinants of housing space and value inequality. We find that accounting for covariates, one percentage point (pp) increase in income inequality (Gini) is associated with up to 0.40pp higher housing value inequality. That relatively high correlation exists at the lowest level of aggregation we study (census tract) and is lower at higher aggregations, such as the municipality level. In other words, on average, dispersion in housing prices is more similar to the income mix in a neighborhood than in a city.

Housing inequality has changed substantially over time. With some caveats to keep in mind, we can measure long-run trends in housing (space) inequality at different aggregations by exploiting the facts that housing is immobile and that we observe dwellings' construction year in the cadastre. We document that housing inequality was much higher than today at the beginning of the 20th Century, with a Gini of 0.4 for Spain. Inequality remained relatively stable during the first few decades of the past century. It decreased rapidly between 1960 and 1980, coinciding with robust economic growth and a boom in construction. It remained relatively stable between 1990 and 2010, with signs of a trend reversal in the 2010s. Despite some notable level differences, most regions

followed the same pattern. Housing value inequality, which we can measure in Catalonia from 2009 to 2019, increased substantially over the 11-year period. In the first few years of the 2010s, it increased due to reductions in dwelling prices, disproportionately affecting the least valuable dwellings, likely due to the burst of a housing bubble and the Great Recession. In the more recent years, it increased due to disproportionate price increases at the top, likely a consequence of the recovery in the economy not being matched by new housing construction. Changes in housing inequality over time appear to be tightly linked to housing supply. Housing demand and macroeconomic conditions during the business cycle appear especially relevant for housing value inequality.

This paper primarily contributes to the work devoted to measuring inequality. A large body of work has employed survey and administrative tax data to measure inequality, contemporaneously and in the long run. Examples include the seminal work by [Piketty and Saez \(2003\)](#) and [Piketty \(2003\)](#), which study income inequality in the US and France since the beginning of the 20th Century, or the more recent work by [Kuhn et al. \(2020\)](#), [Blanco et al. \(2021\)](#), [Zucman \(2019\)](#), which attempt to measure wealth and its long-term dynamics from a more international perspective.

Our work focuses on housing, a multifaceted good that provides shelter, gives access to local labor markets and amenities, and is, at the same time, a source of income and an asset. Despite the role of housing as a key driver of income and, primarily, wealth inequality has been acknowledged in the literature ([Dustmann et al. 2022](#), [Kuhn et al. 2020](#), [Martínez-Toledano 2023](#)), there are relatively few works explicitly devoted to documenting housing disparities in their many different forms. The most notable exceptions and closest papers to ours are [Aladangady et al. \(2017\)](#), which measures housing value inequality in the United States since 1930 using census data, and [Domènech-Arumí and Gobbi \(2025\)](#) and [Domènech-Arumí et al. \(2025\)](#), two companion papers that describe housing inequality in Belgium and the United States. Our work differs from [Aladangady et al. \(2017\)](#) in the type of data used — cadastral data on all dwellings instead of a sample from

the Census —, in the outcomes studied — we look at the full distributions of housing values and space —, and in the spatial scope — we analyze housing inequality at all existing levels of aggregation. In addition, we make all estimates publicly available in the framework of the World Housing Inequality Database (WHID), a new resource that will allow researchers and policymakers to study several questions related to housing from a national or international — all estimates in the WHID will be produced using comparable data and methodologies —, perspective. The WHID database released alongside this paper constitutes a primary contribution of our work.

Our paper emphasizes the importance of granularity in measurement. This is a key distinction with the extensive work on income and wealth inequality, which has predominantly focused on measuring trends at the country level and has prioritized the international comparison of those ([Alvaredo et al. 2013; 2018](#), [Chancel et al. 2022](#)). That distinction is relevant because we uncover substantial geographic heterogeneity in housing inequality estimates which, in light of the extensive work showing that neighborhoods — or, more generally, place — are crucial inputs explaining inequality and social mobility dynamics ([Chetty et al. 2014a;b](#), [Chyn and Katz 2021](#), [Oreopoulos 2003](#)), suggests that characterizing “place” at a more disaggregated level could be essential to studying many different outcomes. Thus, our work also contributes to the neighborhoods literature by providing new data tools to characterize places.

2 Data

2.1 Cadastre and real estate transactions

Cadastre. The Spanish Cadastre (*catastro*) is a public registry provided and maintained by the Ministry of the Treasury for taxation purposes. It contains information on the universe of real estate in the country, including size, construction year, and location of all dwellings. Our data provides a snapshot of all real estate in Spain as of late 2023 and

includes information on approximately 25.5 million dwellings.

The public version of the data we used in this paper does not contain information on dwellings' assessed or market value and ownership, which precludes us from discussing wealth inequality and forces us to complement it with transaction data to discuss value inequality.

Real Estate Transactions. The real estate transactions data come from the Catalan Tax Authority (*Agència Tributaria de Catalunya*, ATC) contain information on the universe of real estate transactions in the region of Catalonia from 2009 to 2019 (over 275,000). We observe the date and sale price of transacted dwellings and, critically, a parcel identifier that facilitates the merging with the cadastre.

2.2 Dwelling value estimations

We use the ATC data to train a predictive model for the log price per square meter. We adopt a random forest algorithm incorporating 14 main predictors identifying dwelling characteristics (e.g., dwelling space and building age). We also include several variables to capture location effects, such as latitude, longitude, and a factor variable identifying the census tract.

Our model yields a prediction error (out-of-bag root mean squared error, OOB-RMSE) of 0.12, indicating that, on average, our value predictions are off by 12% — a figure in line with existing research exploiting similar applications ([Čeh et al. 2018](#), [Mullainathan and Spiess 2017](#)). The model indicates that location is the most relevant factor as the dwelling's census tract, latitude, and longitude arise as the top predictors of a dwelling price per square meter.

We use the trained model to impute a dwelling price to each of the over 3.7 million dwellings in Catalonia for each year between 2009 and 2019.¹

¹See Appendix B for more details about the data, cleaning process, and price prediction process.

3 Housing inequality in Spain

3.1 Measuring inequality

Our discussion is primarily based on the Gini index, chosen for its widespread use in academia, but we complement our analysis with alternative metrics, such as the share of space or value accrued to the top of the distribution. The publicly available dataset released alongside this paper contains several additional measures characterizing different features of the housing space and value distributions.

3.2 Most recent inequality estimates

Housing space inequality. Figure 2 and Table 1 show the geographic variation in housing space inequality estimates across different levels of aggregation as of 2023. These include regions (*comunidades autónomas*, NUTS-2), provinces (*provincias*, NUTS-3), municipalities (*municipios*), and census tracts (*secciones censales*). Table 1 reports the numerical estimates at the country and region level and in the top 12 largest commuting zones (Functional Urban Area, FUA). The table also reports additional inequality metrics (e.g., the share of space accrued to the top 1% of dwellings) and details about specific moments of the distribution (e.g., median housing space). These figures and tables highlight two main results.

First, housing space inequality in Spain is relatively high (Gini of 0.31). This is a relatively large value when compared to Belgium (Domènech-Arumí and Gobbi (2025) estimate a Gini of 0.24) but low compared to the US (Domènech-Arumí et al. (2025) estimate a Gini of 0.35).² Remarkably, our results closely align with those from the OECD concerning income inequality, which estimate after-tax income Ginis (in 2022) of 0.31, 0.25, and 0.4 for Spain, Belgium, and the US, respectively OECD (2022b). These “intermediate” housing inequality values are also reflected in the alternative measures included in Table

²There are currently no other housing inequality estimates against which to benchmark.

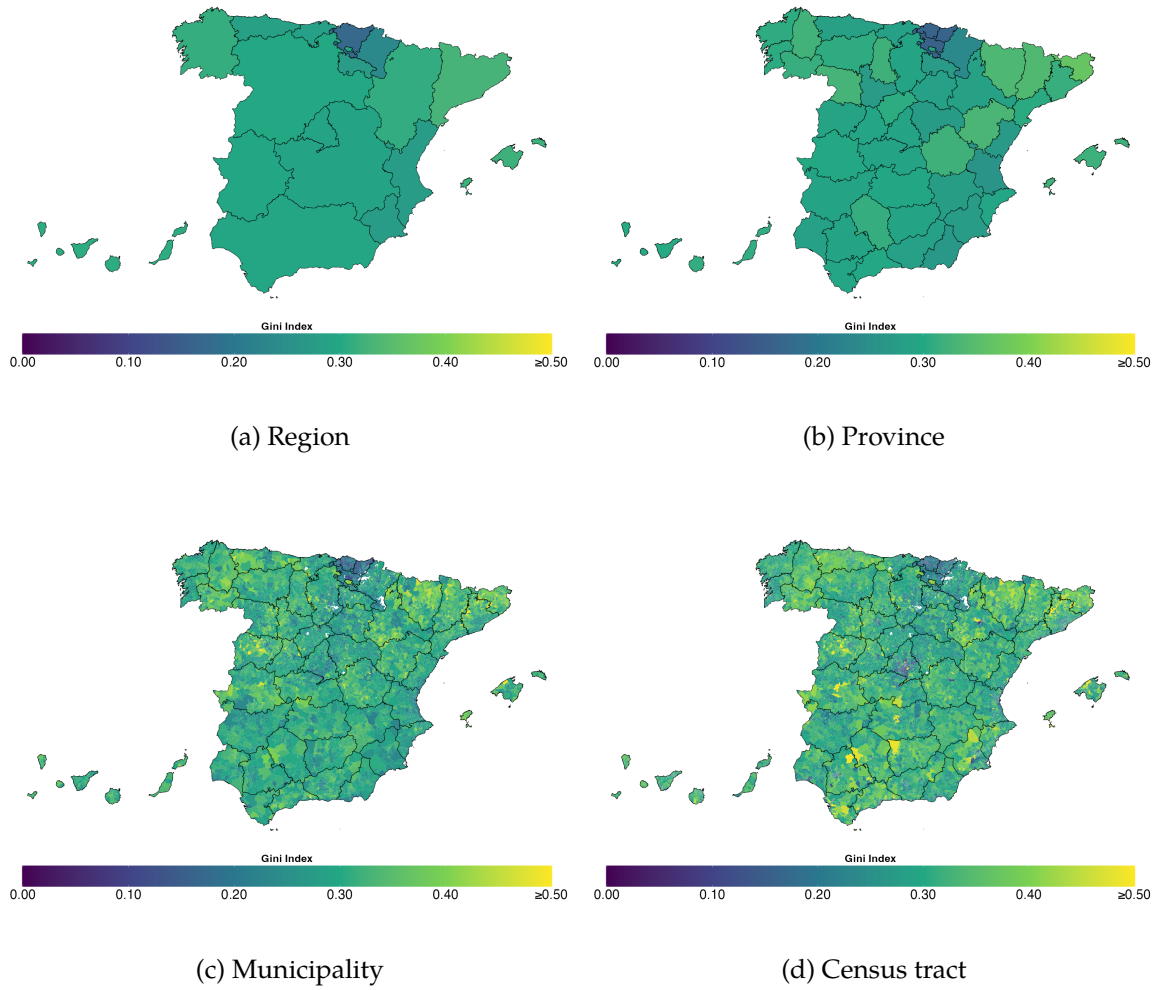


Figure 1: Housing space inequality in Spain

Notes: Each map illustrates housing space (square meters) inequality (Gini) at a given subnational level in 2023. Darker colors describe geographic units with lower levels of inequality. Brighter colors denote higher levels of inequality. Housing space inequality in Spain is 0.307.

1. For example, the 1% most spacious dwellings in Spain contain 5.8% of the total dwelling space in the country. These figures are 3.1% for Belgium and 10.4% for the US. This finding is consistent with a broader literature on income and wealth, which characterizes Spain as having intermediate inequality levels among advanced economies, significantly below those of the United States ([Alvaredo and Saez 2009](#), [Anghel et al. 2018](#)).

Second, there is substantial within-country heterogeneity. Within regions, inequality estimates range between 0.17 (Basque Country) and 0.33 (in Catalonia). At the province

Table 1: Housing space inequality in Spain (2023)

	Gini	Top 1	Top 10	Bottom 50	P99	P90	P50	P10
A. Country								
Spain	0.308	0.058	0.256	0.295	593	270	137	74
B. Region								
Andalucía	0.295	0.057	0.249	0.305	558	257	135	75
Aragón	0.31	0.061	0.264	0.299	695	302	152	85
Canarias	0.306	0.054	0.249	0.295	534	263	131	70
Cantabria	0.284	0.051	0.25	0.315	579	267	136	83
Castilla y León	0.301	0.052	0.252	0.302	654	313	155	87
Castilla-La Mancha	0.293	0.053	0.246	0.307	674	324	166	94
Catalunya	0.328	0.071	0.276	0.285	665	266	130	72
Ceuta	0.315	0.092	0.262	0.292	533	201	122	60
Comunidad de Madrid	0.289	0.057	0.24	0.304	475	232	125	69
Comunitat Valenciana	0.276	0.05	0.235	0.316	519	250	139	78
Extremadura	0.296	0.051	0.243	0.303	623	311	158	86
Galicia	0.312	0.05	0.253	0.294	727	362	172	92
Illes Balears	0.32	0.06	0.265	0.29	688	305	148	79
La Rioja	0.273	0.056	0.241	0.323	645	294	164	98
Melilla	0.302	0.067	0.252	0.301	602	243	132	70
Navarra	0.236	0.042	0.215	0.355	317	172	101	65
País Vasco	0.17	0.031	0.175	0.385	200	117	83	58
Principado de Asturias	0.283	0.056	0.245	0.315	543	232	133	73
Región de Murcia	0.276	0.052	0.234	0.317	545	260	147	81
C. Commuting Zone (Top 12)								
Alicante	0.252	0.046	0.214	0.328	437	227	138	78
Barcelona	0.305	0.072	0.261	0.299	530	233	119	71
Bilbao	0.184	0.038	0.19	0.378	234	122	83	59
Córdoba	0.306	0.06	0.257	0.296	562	245	130	68
Las Palmas	0.301	0.052	0.245	0.298	508	253	130	70
Madrid	0.289	0.056	0.24	0.304	478	234	126	70
Murcia	0.243	0.048	0.219	0.34	528	263	158	98
Málaga	0.276	0.053	0.232	0.313	471	223	129	71
Palma de Mallorca	0.303	0.06	0.257	0.303	668	291	149	84
Sevilla	0.282	0.055	0.24	0.312	484	229	126	72
Valencia	0.239	0.045	0.215	0.34	462	232	142	86
Zaragoza	0.247	0.056	0.222	0.335	464	222	139	83

Notes: *Gini* is the Gini index. *Top 1* and *Top 10* measure the share of housing space that accrues to the top 1 or 10% most spacious dwellings in the specific geography, respectively. *Bottom 50* measure the share of housing space accrued to the 50% least spacious dwellings. *P99*, *P90*, *P50*, and *P10* indicate the space of the dwelling in the 99th, 90th, 50th, and 10th of the dwelling space distribution in the specific geography, respectively. Housing space is measured in square meters. *Communing Zones* proxied by Functional Urban Areas (FUAs) (only the 12 largest cities are included in this table).

level, estimates range between 0.16 (Álaba) and 0.37 (Girona). At the commuting zone level, they range between 0.16 (San Sebastián) and 0.36 (Girona). And there is even larger variation within cities (census tract level), as illustrated by Figure 1 or Appendix Figure A1, which shows significant variation within Madrid and Barcelona (the two largest cities).

As indicated by Table 1, relatively high values of inequality are typically associated with small housing sizes at the bottom (dwellings of approximately 70 square meters in the 10th percentile).³ As indicated by Panel A in Table A1, nationwide inequality (as measured by the Theil index) is overwhelmingly driven by variation within provinces. Within cities (FUAs), inequality within census tracts explains most (about two-thirds) of the variation, with significant heterogeneity across cities. The bottom two panels in Appendix Figure A2 plot median dwelling sizes across geographies and illustrate this within-province variation well. We will investigate and discuss the determinants of housing space inequality in more detail in Section 4.

Housing value inequality (in Catalonia). Housing value inequality in Catalonia, as measured by the Gini index, is 0.37. This is above the estimate for housing space in the region (0.33). As with space, Figure 2 and Table 2 document significant geographic heterogeneity across administrative levels. For example, Barcelona and Lleida stand out as the most unequal units at the province level (Gini of 0.36), and Girona, which was the most unequal province in Spain in housing space, emerges as the least unequal in the region (Gini of 0.31).

Barcelona and Lleida are the most unequal provinces, for very different reasons. In Barcelona, inequality is high because dwellings are very valuable, especially at the top. For instance, a dwelling in the 99th percentile of the distribution costs over a million euros (in 2019), which is almost double the price of a dwelling at the same point of the distribution in Girona, the second most expensive province. In contrast, the top dwellings

³For context, a dwelling in the 10th percentile contains about 90 square meters in Belgium, and 74 in the US.

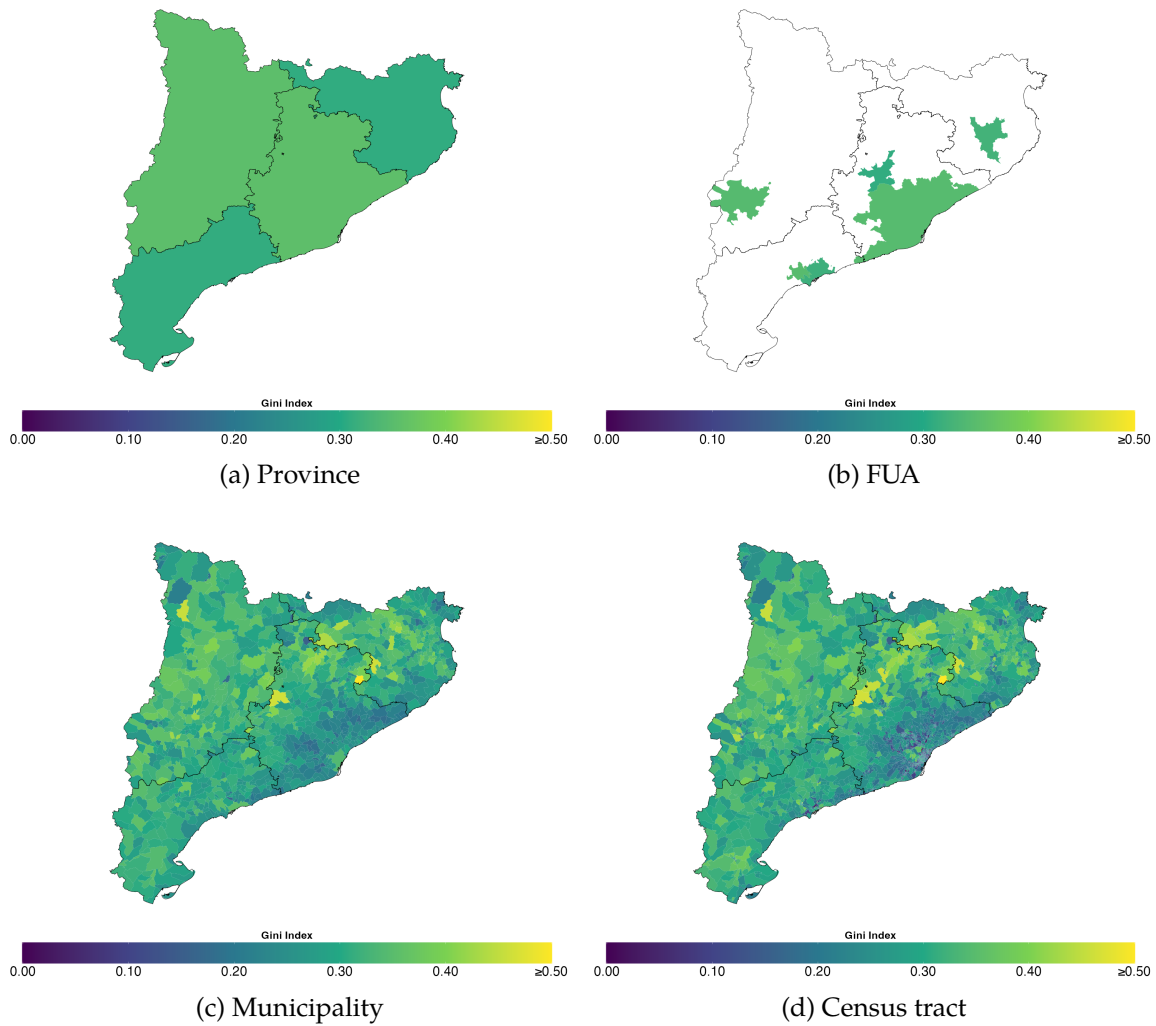


Figure 2: Housing value inequality in Catalonia

Notes: Each map illustrates housing value (euros) inequality at a given subregional level in 2019. Darker colors describe geographic units with lower levels of inequality. Brighter colors denote higher levels of inequality. Housing value inequality (Gini) in Catalonia was 0.29 in 2019.

in Lleida sell for less than half a million euros, and those at the bottom are inexpensive (a dwelling at the bottom decile sold for 42,000 euros in 2019), thus widening the support of the distribution from below and driving inequality up. Remarkably, these differences in value arise despite the fact that dwellings in Lleida are significantly larger than those in Barcelona province. For instance, the median dwelling has 175 square meters in Lleida and only 124 in Barcelona. However, the median dwelling's price per square meter (in 2019) was 1,600 euros in Barcelona and only 540 euros in Lleida (See Appendix Table A2

Table 2: Housing value inequality in Catalonia (2019)

	Gini	Top 1	Top 10	Bottom 50	P99	P90	P50	P10
A. Region								
Catalonia	0.373	0.065	0.288	0.251	899	394	169	76
B. Province								
Barcelona	0.355	0.063	0.279	0.263	1002	456	204	97
Girona	0.309	0.052	0.245	0.289	559	277	139	73
Lleida	0.355	0.058	0.269	0.26	492	225	104	44
Tarragona	0.311	0.049	0.242	0.289	413	219	110	53
C. Commuting Zone								
Barcelona	0.346	0.062	0.275	0.269	1040	475	217	106
Girona	0.325	0.051	0.246	0.276	497	291	131	70
Lleida	0.346	0.056	0.25	0.259	422	218	108	42
Manresa	0.309	0.052	0.241	0.289	428	229	115	56
Reus	0.347	0.053	0.252	0.256	458	256	121	50
Tarragona	0.318	0.049	0.245	0.281	455	247	117	58

Notes: *Gini* is the Gini index. *Top 1* and *Top 10* measure the share of housing value that accrues to the top 1 or 10% most valuable dwellings in the specific geography, respectively. *Bottom 50* measure the share of housing value accrued to the 50% least valuable dwellings. *P99*, *P90*, *P50*, and *P10* indicate the value of the dwelling in the 99th, 90th, 50th, and 10th of the dwelling value distribution in the specific geography, respectively. Housing value is measured in 1000s of euros. *Commuting Zones* proxied by Functional Urban Areas (FUAs) (only the largest cities are included in this table).

and Appendix Figures A3 and A4).⁴

Finally, Panel B in Table A1 shows that overall inequality in the region (as measured by the Theil index) is overwhelmingly driven by variation within provinces. Within cities, variation within census tracts explains about two-thirds of inequality levels, a result similar to what we found for space. The notable exception is Barcelona, where over 50% of the variation arises from the between-tracts component – indicating that housing value inequality in Barcelona is predominantly driven by large differences between neighborhoods.

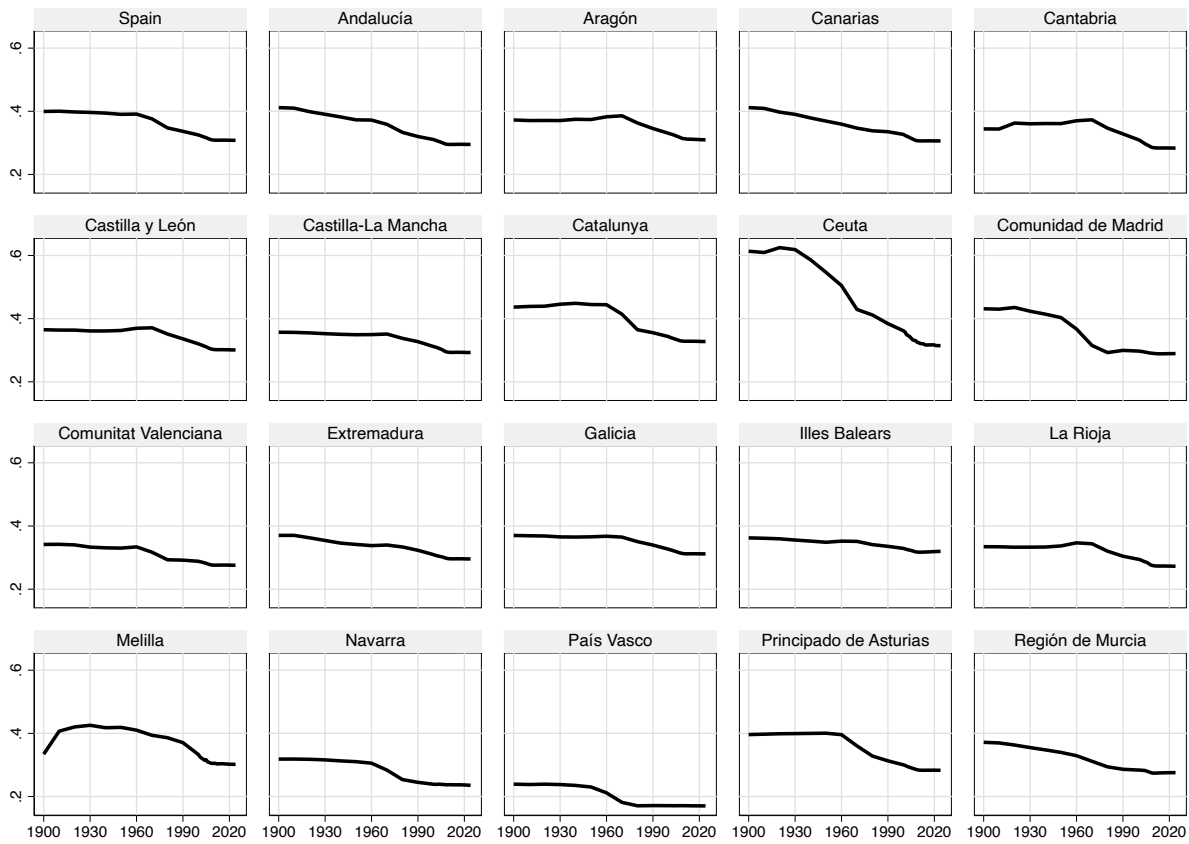


Figure 3: Evolution of housing space inequality (Gini) across regions

Notes: Housing space is measured in square meters. The top-left panel describes the evolution of housing space inequality (Gini) for Spain. The rest of the panels report the evolution of housing space inequality at the regional level.

3.3 Housing inequality over time

Housing space. To construct long-run inequality estimates, we exploit the panel dimension of our cross-sectional data, using each dwelling's construction year to reconstruct the housing stock at different points in time. The primary caveat to this method is that we do not observe dwellings exiting the housing stock (e.g., demolitions), which requires caution when interpreting some of the results characterizing the very long run.

⁴Note that all value estimates discussed refer to 2019, the most recent year available in the transactions data. Housing prices have increased substantially since the COVID-19 pandemic, particularly in Barcelona, so housing value inequality in 2025 could be significantly different.

For Spain, this concern could be particularly relevant for estimates in specific areas (such as Barcelona, Madrid, or Gernika in the Basque Country) before 1939 due to bombings during the Spanish Civil War (1936-39).⁵

With these caveats in mind, we find that housing space inequality in Spain has declined over the 20th Century, particularly after 1950-60. The decline halted (and reverted slightly) in the first two decades of the 21st Century. This is well illustrated in the top-left panel in Figure 3, which shows the evolution of housing space inequality (Gini) at the country level. The remaining panels of the figure indicate that almost all the regions followed the same trend, with inequality declining over time and an acceleration in the 1950s-60s.

The figure suggests there is a certain consistency in the ranking of regions. For instance, Catalonia consistently rises as the most unequal region, with a Gini above 0.4 for most of the past century. Conversely, Navarra and Basque Country are always at the bottom, with Ginis generally below 0.3 and around 0.2 towards the end of the period.

The noticeable exceptions (which we want to caveat due to severe selection concerns) are Ceuta and Melilla, the two autonomous cities located in the African continent and considered as regions in the eyes of the administration. For Ceuta, we estimate a Gini above 0.6 in the first decades of the 20th Century. Records suggest that approximately 13,000 people lived in Ceuta in 1900, but our methodology only identifies 64 dwellings in that city in that year. The real figure was likely to be closer to 2,000 dwellings, suggesting that we only capture 3% of all dwellings present at the time (nationwide, we estimate our data captures 33% of all the dwellings present in 1900). The numbers are similar for Melilla, with a population close to 10,000 and only 83 dwellings based on cadastral data. Because of the small samples (in absolute and relative terms), our estimates for the first decades of the Century in these two autonomous cities are likely to misrepresent the actual dwelling stock significantly.⁶

⁵See Section B.1.3 in the Appendix for a discussion about the quality of our long-run estimates.

⁶We decided to report the long-run estimates for Ceuta and Melilla, even though we believe they especially biased, for transparency.

A closer look at the entire dwelling space distribution reveals the source of this trend. As shown in Appendix Figure A5, the long-run decline in the Gini was predominantly driven by a compression at the top of the distribution, specifically, a sharp reduction in the size of the most spacious dwellings that accelerated after 1960. For instance, in Catalonia, a dwelling in the 90th percentile contained 433m² in 1950, 391m² in 1960, and 302m² in 1970. The same pattern is present in most regions, even in Navarra and the Basque Country, two regions with a consistently compressed dwelling space distribution. The sharp change in the space dwelling distribution post-1960 coincides with a period in which the Spanish economy started a liberalization process after 20 years of stagnation and autarky. We further unpack the temporal variation and expand this discussion in Section 4.2.

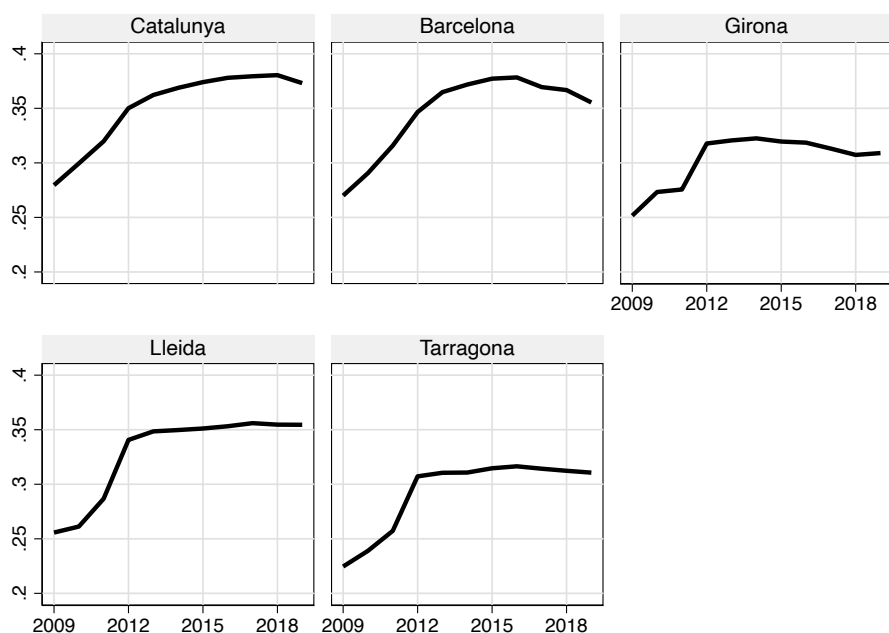


Figure 4: Evolution of housing value inequality (Gini) across provinces

Notes: Housing value is measured in euros. The top-left panel describes the evolution of housing value inequality (Gini) for Catalonia. The rest of the panels capture the evolution of housing value inequality at the province level.

Housing value (in Catalonia). Figure 4 illustrates recent trends in housing value inequality (Gini) in Catalonia (top-left) and its four provinces. Our discussion concerning housing value inequality trends is restricted to Catalonia in 2009-19 due to the nature of our housing transactions data. This shorter time span precludes us from forming a long-term perspective, but it has the advantage of reduced concerns about housing outflows (e.g., demolitions) not being observed in the data.

Housing value inequality has increased significantly in Catalonia since 2009, with most of the increase occurring in the first three years (2009-12). Over this short period, housing value inequality (Gini) increased from 0.28 to 0.35 (25%), with comparable increases occurring across provinces. Inequality kept growing in the latter period (2013-19), but at a much slower pace. We estimate a housing value Gini of 0.37 in 2019. It is worth inspecting the dwelling value distribution and macroeconomic environment to interpret these trends.

Appendix Figure A6 reveals that the increase in the Gini in the first few years is predominantly driven by a substantial decrease in the value of dwellings at the bottom of the distribution. For instance, a dwelling in the 10th percentile was worth approximately 107 thousand euros in 2009 but only 70 thousand in 2012 (35% decrease). Dwellings at the top also experienced a reduction in value, but the price correction was milder. Over the same period, we estimate that the price of a dwelling in the 90th percentile decreased from 340 to 306 thousand euros (10% decrease). This period coincides with the burst of the Spanish housing bubble (2008) and the worst years of the Great Recession (e.g., nationwide unemployment peaked at 26% in 2013). These adverse macroeconomic conditions are likely to have significantly reduced the demand for housing, especially at the bottom, which resulted in the widening in the housing value distribution and, therefore, in an increase in the Gini. This hypothesis is supported by [Anghel et al. \(2018\)](#), which documents a sizable decrease in expenditure on consumer durables among low-income households during the recession.

The trend continued at a reduced pace in 2013-15, with dwellings at the bottom decile reaching the lowest price tag at 61 thousand euros (2014) and those at the top a floor of 291 thousand euros (2013). These years coincided with the start of a mild recovery of the Spanish economy.

In the most recent sample period (2016-19), housing inequality continued increasing at a reduced pace, but as a consequence of a widening distribution from the top. Dwellings at the bottom decile had only regained half of their lost value by 2019 (a dwelling sold for 76 thousand euros, still almost 30% below their 2009 price). Those at the 90th percentile completely erased the losses and increased their valuation to 394 thousand euros by 2019, a 16% increase relative to 2009.

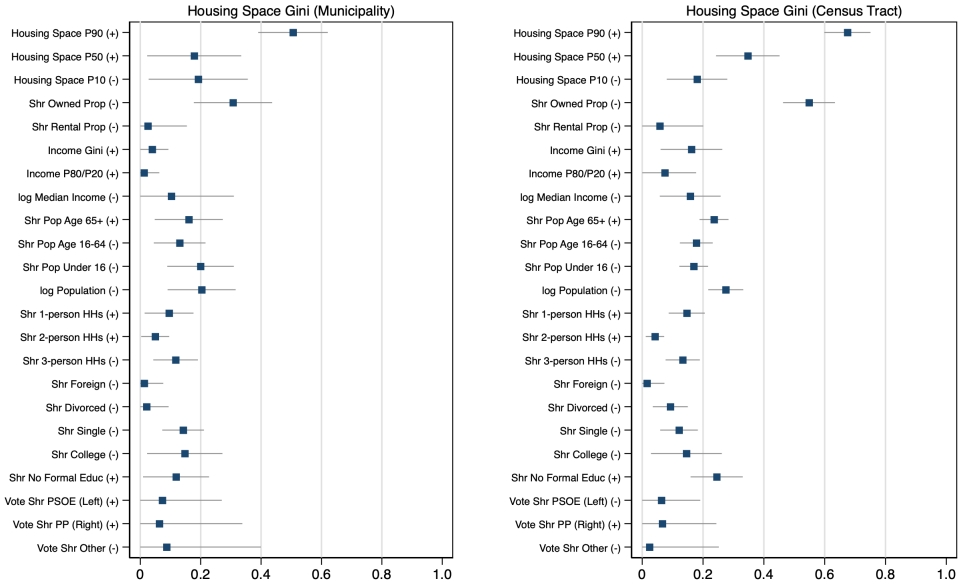
Appendix Figure A6 highlights that Barcelona is the main responsible for the recent increase in value inequality driven by price increases at the top.

Our time series stops in 2019, but it is well known that housing prices have kept rising rapidly since then, especially after the COVID-19 pandemic. It is unclear how housing value inequality (e.g., the Gini) has changed since 2019 because anecdotal evidence indicates that dwellings at all points of the distribution have increased their valuations, especially in Barcelona, where exploding demand and stagnated supply have brought housing affordability at the forefront of the public debate.

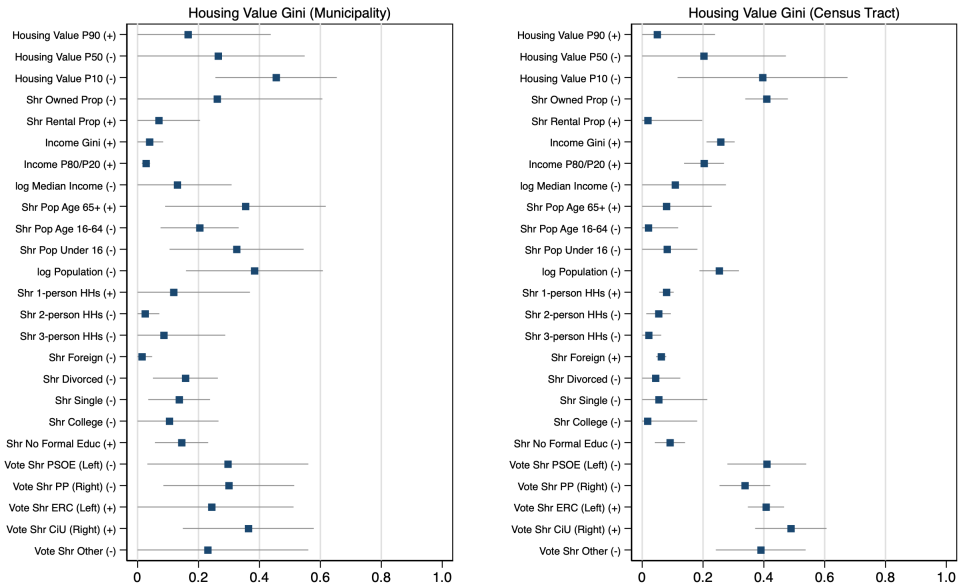
4 The determinants of housing inequality

4.1 What explains housing inequality today?

Correlations. We first examine the simple correlations between our inequality (Gini) estimates and several demographic and socioeconomic covariates obtained from the Spanish Census and other sources. Figure 5 reports correlations at the municipality and



A. Housing Space



B. Housing Value

Figure 5: The determinants of contemporaneous housing inequality

Notes: This figure reports the raw correlations between housing inequality estimates (space in Panel A or value in Panel B) and several covariates at the municipality or census tract level. The + and – signs next to a covariate label denote a positive and negative correlation, respectively. Housing space and value inequality (Gini) estimates are measured in 2019. *Housing Value* or *Space* P10, P50, and P90 are constructed from the cadastre and measured in 2019. *Income Gini*, *Income P80/P20* and *log Median Income* are retrieved from INE's *Atlas de la Renta* and measured in 2019. *Vote Shr PSOE (Left)*, *Vote Shr PP (Right)*, *Vote Shr ERC (Left)*, and *Vote Shr CiU (Right)* are the share of votes received by the PSOE, PP, ERC, and CiU in the 2011 National Election, respectively. ERC and CiU are two parties only relevant in Catalonia. *Vote Shr Other* is the vote share obtained by all parties except the PSOE and PP (top panel) or PSOE, PP, ERC, and CiU (bottom panel) in the same election. The rest of the covariates are obtained from the Census and measured in 2011.

census tract level.⁷

Panel A reveals that housing space inequality is significantly higher in municipalities and tracts with larger dwellings and lower in places with smaller dwellings. That is a mechanical relationship. The panel also reveals that places with higher ownership rates have more homogeneous dwellings and, therefore, lower inequality levels (correlation of 0.3 at the municipality level and 0.5 at the tract level). The figure also indicates a mild correlation (approximately 0.2) with several demographic characteristics. Inequality is significantly higher in places with older (65+) individuals and lower in places with more working-age, younger, and single individuals. It is also higher in places with more individuals without formal education and higher income inequality, even though these correlations are only significant when zooming in at the tract level, which generally yields larger and more precise correlations.

Panel B reports the correlations with housing value inequality (in Catalonia). It shows that places with higher dwelling valuations at the bottom are significantly less unequal (correlation of -0.5), which is, again, somewhat mechanical — even though the positive correlations with dwelling valuations at the top are not significant. The panel also reveals strong correlations (about 0.4 for estimates at the municipality level) with the population's demographic characteristics, with higher inequality in areas with older individuals and lower inequality in places with younger populations. In contrast with the results for space, both figures in the panel show relatively strong (0.3-0.5) correlations between housing value inequality and electoral outcomes, although perhaps not in the expected direction. We find that places exhibiting higher electoral support for the two main Spanish parties (PSOE in the center-left and PP in the center-right) are less unequal. Places with higher electoral support for the two main Catalan parties (ERC in the center-left and CiU in the center-right) are more unequal. In other words, the dividing

⁷The census tract is the lowest existing level of aggregation. The municipality is the highest meaningful level of aggregation for this exercise (given that value estimates are only available in one region with four provinces).

gradient appears to be in the national rather than ideological spectrum. Finally, the figure indicates a mild correlation (approximately 0.25) with income inequality (Gini), which is only significant at the census tract level.

Table 3: The determinants of housing inequality

	Housing Space Inequality (Gini)				Housing Value Inequality (Gini)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Income Gini	0.021 (0.025)	0.072** (0.025)	0.464*** (0.016)	0.416*** (0.018)	0.031 (0.049)	0.027 (0.054)	0.395*** (0.031)	0.342*** (0.034)
log Med Income		-0.012 (0.006)		-0.023*** (0.003)		0.007 (0.014)		-0.067*** (0.007)
log Population		-0.009*** (0.001)		-0.012*** (0.001)		-0.004** (0.002)		-0.009*** (0.002)
Shr Pop Age 16-64		0.060 (0.033)		-0.006 (0.011)		0.084 (0.067)		0.007 (0.024)
Shr Pop Age 65+		0.061* (0.025)		-0.047*** (0.011)		0.139* (0.055)		-0.128*** (0.022)
Shr No Formal Educ		0.068*** (0.018)		0.377*** (0.013)		0.179** (0.059)		0.190*** (0.033)
Shr Foreign		0.010 (0.010)		-0.082*** (0.006)		0.052* (0.023)		-0.052*** (0.011)
Shr Owned Prop		-0.030*** (0.006)		-0.191*** (0.004)		0.002 (0.014)		-0.055*** (0.008)
Vote Shr PSOE		-0.048*** (0.010)		-0.133*** (0.008)		0.040 (0.056)		0.043 (0.038)
Vote Shr PP		-0.042*** (0.009)		-0.026*** (0.007)		-0.034 (0.050)		-0.046 (0.031)
Vote Shr ERC						0.080 (0.060)		0.145** (0.044)
Vote Shr CiU						0.095* (0.044)		0.205*** (0.027)
DV Mean	0.301	0.301	0.220	0.220	0.290	0.290	0.225	0.225
DV SD	0.055	0.055	0.094	0.094	0.049	0.049	0.076	0.076
Adj-R ²	-0.000	0.510	0.036	0.437	-0.001	0.282	0.043	0.429
N	4195	4195	21302	21302	875	875	4001	4001
Province FE		X		X		X		X
Aggregation	Mun.	Mun.	Tract	Tract	Mun.	Mun.	Tract	Tract

Notes: *Income Gini*, *Income P80/P20* and *log Median Income* are retrieved from INE's *Atlas de la Renta* and measured in 2019. *Shr Pop Age 16-64*, *Shr Pop Age 65+*, *Shr No Formal Educ*, *Shr Foreign*, and *Shr Owned Prop* are retrieved from the 2011 Census. *Vote Shr PSOE*, *Vote Shr PP*, *Vote Shr ERC*, and *Vote Shr CiU* are the share of votes received by the PSOE, PP, ERC, and CiU in the 2011 National Election, respectively. All independent variables are demeaned. Heteroskedasticity-robust standard errors in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Regression analysis. We complement the previous correlation exercise by estimating the

following model:

$$HousIneq_i = \alpha + \beta IncIneq_i + \gamma X_i' + \epsilon_i \quad (1)$$

Where *HousIneq* is the 2019 housing space or value inequality (Gini) estimated in place *i*, which could be a municipality or census tract (as in Figure 5). *IncIneq* is the 2019 income inequality estimate in *i*. *X* is a vector containing several demographic and socioeconomic covariates that displayed a high correlation with housing inequality in Figure 5, with most of them coming from the 2011 Census and the 2011 national election. *X* also includes province fixed effects. Table 3 shows the results.

No covariate is strongly associated with housing inequality at the municipality level. Even though there is a relatively high association between housing space and income inequality at the municipality level after accounting for covariates (Column 2), correlations are far weaker than in the census tract-level regressions.

Income inequality (Gini) is one of the most important determinants of housing space and value inequality at the census tract level. A point increase in income inequality is associated with a 0.4-0.46 point increase in housing space inequality (Columns 3-4) and a 0.35-0.40 point increase in housing value inequality (Columns 7-8). The remaining covariates exhibit the same correlation signs as in Figure 5 and are statistically significant for the most part.

Overall, income inequality is highly correlated with housing (space and value) inequality. However, the correlations in the Spanish setting are below what Domènech-Arú and Gobbi (2025) find for Belgium. In that paper, the raw correlations between income and housing inequality ranged from 0.1 to 0.4 and from 0.4 to 0.8 for space and value inequality, respectively, depending on the level of aggregation studied. Housing value is more correlated with income than housing space is in both settings, but the correlation is stronger in the Belgian context.

On a related note, based on the results reported in Appendix Table A1, the higher correlations we observed between income and housing inequality at the census tract

level are explained by the fact that inequality at the city level appears to be mostly (about two-thirds) driven by variation within (as opposed to between) census tracts. The notable exception is Barcelona (Panel B), where the variation between tracts is much more relevant. In other words, at least in the largest city in the region, overall inequality is predominantly driven by variation in housing values between (rather than within) tracts (a proxy for neighborhoods), which may indicate that, within a tract, variation in the income of residents could be lower. Consistent with this idea, we find that the raw correlation between income and housing value inequality at the tract level increases from 0.26 to 0.33 (27% increase) after excluding Barcelona.

4.2 What explains the change in housing inequality over time?

Changes in the space distribution of new inflows. Changes in the space distribution of new dwelling inflows are crucial to explaining changes in housing space inequality over time. Figure 6 precisely illustrates this point by showing housing space inequality at the cohort level, where we define a cohort as all dwellings entering the stock in the previous decade. For instance, the figure reveals a much more unequal space distribution among dwellings built between 1901 and 1910 (1919 cohort, Gini of 0.4) than those built between 2001 and 2010 (2010 cohort, Gini of 0.24). Because it takes significant time to change the distribution in the stock of a good as durable as housing, looking at the space distribution of housing inflows is essential to nuance and interpret Figure 3.

New housing inflows were significantly more unequal at the beginning of the 20th Century. Inequality within cohorts declined slightly in the decades leading up to 1960 (e.g., Gini of 0.36 for Spain), when that trend accelerated rapidly for two decades (Gini of 0.28). Inequality in the subsequent cohorts kept decreasing gradually until 2010 (Gini of 0.24), and the trend reversed course in the latest cohort in the series (2020), with an uptick in inequality (Gini of 0.28).

Most regions have followed the same trends. The most notable exceptions are

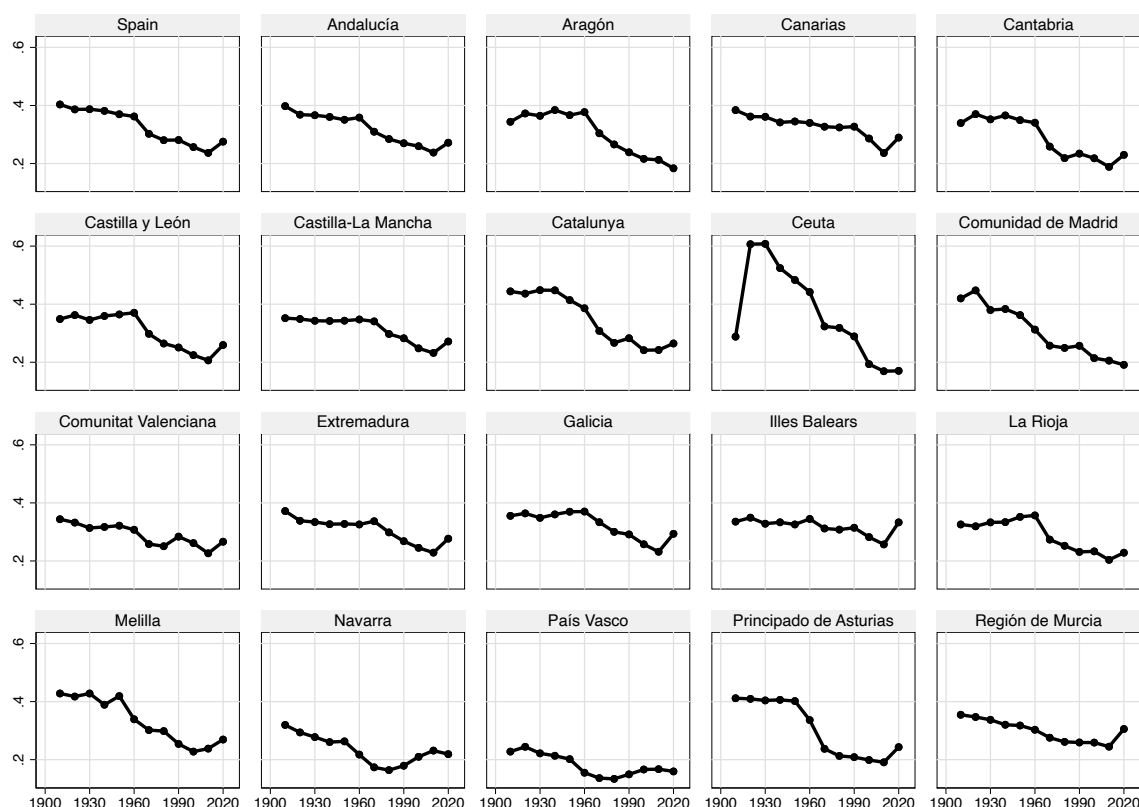


Figure 6: Evolution of housing space inequality (Gini) by decade-cohort across regions

Notes: Housing space is measured in square meters. The top-left panel describes the evolution of housing space inequality (Gini) by decade-cohort for Spain. The rest of the panels capture the evolution of the same outcome for the regions. Estimates are constructed based on the dwellings entering the housing stock in the previous decade (e.g., the observation for 1910 is based on the dwellings constructed between 1901 and 1910).

Catalonia, Valencia, Navarra, and the Basque Country, which experienced slight increases in inequality within cohorts starting as early as 1980. Ceuta also appears to be an outlier in the figure, as dwellings constructed over the first decades of the 20th Century appear to have been extremely unequal.⁸

The reduction in inequality in the first few cohorts in the 20th Century was driven by a significant decrease in the size of dwellings at the top, as revealed by Appendix Figure A7. For example, the size of a dwelling in the 90th percentile of the distribution in the 1910

⁸As previously discussed in Section 3 and in Appendix B, we want to caution the reader of not overinterpreting the estimates for Ceuta (and Melilla) in the first decades of the 20th Century due to sample selection concerns.

cohort was 396m². A dwelling in the same percentile contained 329m² in the 1950 cohort, 261m² in the 1960 cohort, and 206m² in the 1970 cohort (the smallest size in this percentile in the series). In contrast, dwellings at the bottom decile contained approximately the same space (about 65m²) throughout the period.

The continued inequality decrease in the second half of the 20th Century is primarily driven by space gains at the bottom. New dwellings at the top regained space in the post-1970 cohorts, with sizes ranging between 250 and 300m². Those at the bottom, however, saw substantial increases, with sizes in the first decile rising from 60m² in the 1960 cohort to 95m² in the 2010 cohort (58% increase). There was a substantial compression in the space distribution from the bottom.

Despite differences in levels, most regions exhibited similar trends. Appendix Figure A7 shows substantial variation in the distribution of housing space by cohort across regions. For instance, the largest newly constructed dwellings in places like Aragón, Cantabria, Castilla y León, or Catalonia were very spacious at the beginning of the past century, with sizes in the top decile nearing 500m². These regions followed the previously described trends and ended the series with sizes close to 250m² in the same decile. In contrast, places like Navarra and, especially, the Basque country always exhibited remarkably equal distributions. In the Basque Country, a dwelling in the top decile in the 1910 cohort only contained 147m². One in the bottom decile contained 52m². 110 years later, dwelling sizes in the 2020 cohort are 116 and 61m² in the top and bottom deciles, respectively.

Size of inflows (housing supply). Reductions in housing space inequality are associated with sizable housing inflows (new construction). Figure 7 shows the evolution in the decennial percentage change in housing inequality (solid-red line) and the decennial percentage change in the housing stock (dashed-blue line). The top-left panel shows that Spain's housing stock grew between 10 and 20% per decade throughout most of the

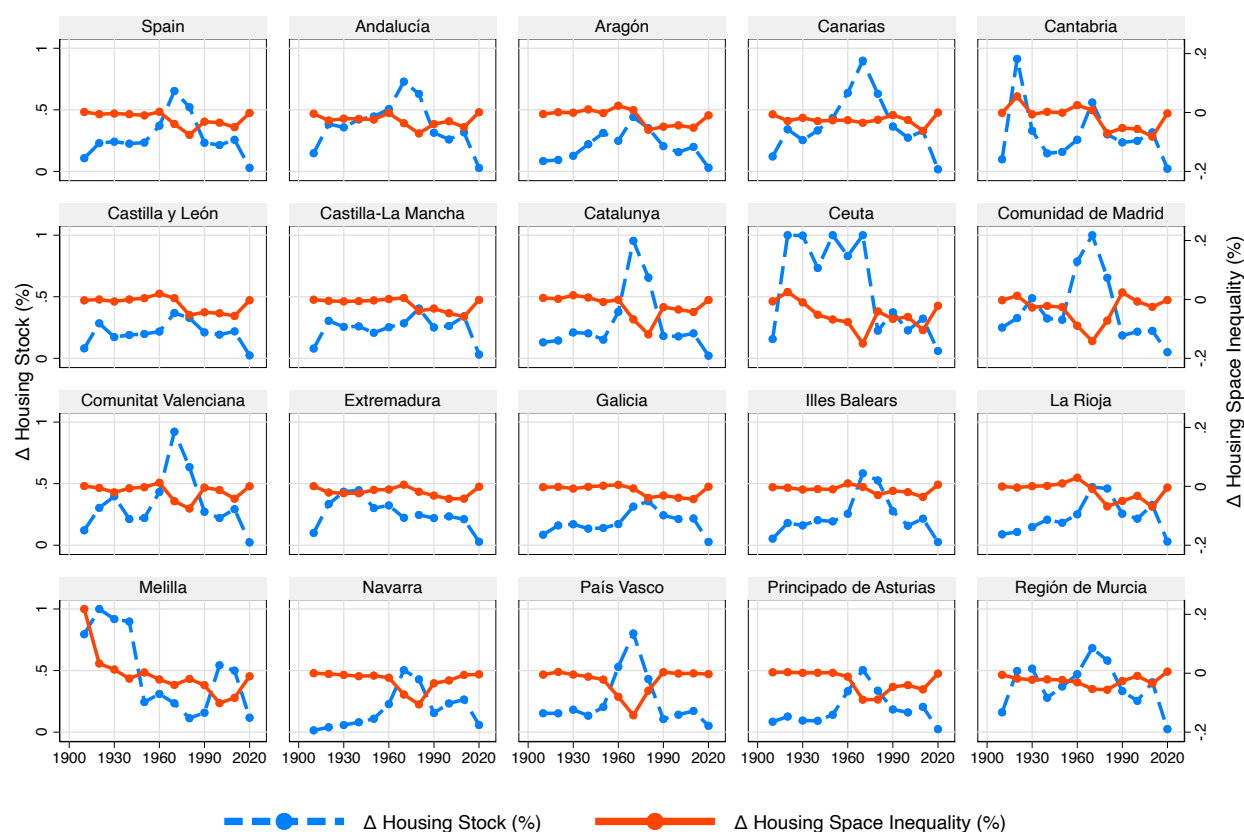


Figure 7: Change in housing inequality and stock over time across regions

Notes: $\Delta Housing\ Stock\ (\%)$ is the decennial percentage change in the housing stock. $\Delta Housing\ Space\ Inequality\ (\%)$ is the decennial percentage change in housing space inequality (Gini). Housing stock growth in the figure is censored at 100% for illustration purposes.

period (1900-2020). This sustained growth coincided with high stability in overall housing inequality, with changes in the Gini nearing zero in most decades.

The period between 1960 and 1980 stands out in the series. These were decades in which Spain's housing stock grew significantly faster, with a boom in construction leading to a 60% growth rate between 1961 and 1970. That coincided with a substantial reduction in housing space inequality. In contrast, the most recent decade in the data (2011-2020) is characterized by anemic growth, with only under 3% of new dwellings added into the stock over the entire decade. That coincided with a recent spike in inequality within that last cohort.

The remaining panels in the Figure reveal similar patterns across all Spanish regions,

with general spikes in new housing inflows in 1960-80 coinciding with decreases in inequality and a recent plummet in new construction coinciding with an uptick in inequality.

Table 4: Changes in housing inequality and housing stock

	Δ Housing Space Inequality								Δ Housing Value Inequality			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Housing Stock	-0.084*** (0.014)	-0.078*** (0.018)	-0.077*** (0.009)	-0.068*** (0.009)	-0.086** (0.027)	-0.082** (0.027)	0.009 (0.021)	0.012 (0.023)	-0.203*** (0.025)	-0.201*** (0.027)	-0.061** (0.022)	-0.060** (0.021)
Δ Population		-0.027 (0.030)		-0.042* (0.020)		-0.003* (0.001)		-0.161 (0.161)		-0.002 (0.004)		-0.002 (0.003)
DV Mean	-0.023	-0.023	-0.020	-0.020	0.001	0.001	0.034	0.034	-0.001	-0.001	0.002	0.002
DV SD	0.031	0.031	0.027	0.027	0.014	0.014	2.839	2.839	0.011	0.011	0.033	0.033
Adj-R ²	0.300	0.301	0.279	0.290	0.033	0.034	-0.000	0.000	0.218	0.217	0.024	0.023
N	218	218	608	608	7433	7433	21851	21851	847	847	2829	2829
Aggregation	Region	Region	Prov.	Prov.	Mun.	Mun.	Tract	Tract	Mun.	Mun.	Tract	Tract

Notes: Δ Housing Space Inequality and Δ Housing Value Inequality are the decennial percentage change in housing space or value inequality (Gini), respectively. Δ Housing Stock is the decennial percentage change in the housing stock. Δ Population is the decennial percentage change in population. Heteroskedasticity-robust standard errors in parenthesis. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

To test this relationship more formally, we move beyond graphical evidence and estimate the following model:

$$\Delta HousIneq_{it} = \alpha + \beta \Delta H_{it} + \gamma \Delta Pop_{it} + \epsilon_{it} \quad (2)$$

Where $\Delta HousIneq_{it}$ is the 10-year change in housing inequality (Gini) in place i during the t decade. ΔH_{it} and ΔPop_{it} are the 10-year change in the housing stock and population in i during the decade t . Table 4 reports the results.

A 10pp increase in the housing stock over a decade is associated with a 0.8pp reduction in housing space inequality (Columns 1-2), which quantitatively confirms what was apparent in Figure 7. The sign and overall magnitude of the association holds when restricting the analysis to the province and municipality levels (Columns 3-6). There is no clear association between new housing inflows and housing space inequality at the census tract level.

The association is robust to accounting for population growth. In other words, places that grow more tend to see milder reductions in housing space inequality (up to 0.4pp for

a 10pp increase in population over a decade), but population changes are not the main driver of housing inequality. Independently of population dynamics, places that build more housing see bigger reductions in inequality (Columns 2, 4, and 6).

Housing supply is especially relevant to understanding the dynamics of housing value inequality. Columns 9-12 explore the relationship between growth in the housing stock from 2010-19 and changes in the housing value Gini index at the municipality (Columns 9-10) and census tract (Columns 11-12) level. Our analysis is restricted to one decade due to the time coverage of the transaction data. Results indicate that a 10pp increase in housing supply is associated with a 0.2pp decrease in housing value inequality at the municipality level and a 0.06pp decrease at the tract level. This association is robust to accounting for population growth.

4.3 Discussion

4.3.1 Housing supply and demand

Housing supply matters. The latest results showcased in Table 4 highlighted a strong negative association between housing supply and housing space and value inequality. We saw that housing space inequality decreased significantly between 1960 and 1980, a period characterized by an unprecedented expansion in housing supply, which appears to be at the core. To add some historical context, the key event that likely triggered the housing expansion was the Stabilization Plan (*Plan de Estabilización*) of 1959, which put an end to 20 years of autarky through a series of market liberalization reforms that brought the “golden age” of the Spanish economy (de la Escosura and Rosés 2009, de la Escosura et al. 2011).

Housing value inequality is even more sensitive to housing supply. The 2010-19 period highlights this perfectly. Note that even though space inequality in new housing inflows increased during the period (Figure 6), with dwellings in the top decile containing more space than those built in previous decades (Figure A7), value inequality

decreased significantly in the places that built more (Table 4). The underlying mechanism is straightforward: housing becomes relatively less scarce in places with larger inflows, which reduces housing prices and compresses the dwelling value distribution (i.e., reducing inequality).

Aggregate demand is also crucial to explaining the dynamics of housing inequality. The 1960s construction boom was a consequence of developers responding to the new free market incentives and economic growth that followed after two decades of failed central planning. The 2009-12 increase in value inequality in Catalonia (Figure 4) is a consequence of a drop in the demand stemming from a severe recession that affected working classes disproportionately (Anghel et al. 2018), which reduced housing demand and, therefore, prices in the lowest segment of the housing market. As with wealth inequality, changes in housing prices during business cycles are crucial determinants to short-run housing value inequality dynamics (Martínez-Toledano 2023).

4.3.2 What is housing inequality?

This final subsection discusses our perspective and interpretation of the housing inequality estimates produced and described in this paper.

Housing space and value inequality literally measure the variation in the distribution of housing sizes and prices. These are important objects to study *per se*. For instance, housing space is a critical variable in the literature on overcrowding, which links a lack of housing space to adverse outcomes, such as lower test scores and reduced well-being, especially among children (Goux and Maurin 2005, Kohl et al. 2024, Lopoo and London 2016, Solari and Mare 2012). It may also be linked to fertility (Kulu and Vikat 2007). On the other hand, housing is an important asset and consumption good, as well as the target of many government policies such as mortgage interest deductions and rental subsidies, which makes keeping track of housing values crucial to measure consumption disparities and evaluate the impact of these policies (Braakmann and McDonald 2020, Damen et al.

2016, Eriksen and Ross 2015, Glaeser and Shapiro 2003).

Housing inequality is also informative about other forms of inequality, which researchers and policymakers may care about. We found a relatively strong correlation between housing value inequality and income inequality, consistent with the findings of Domènech-Arú and Gobbi (2025) for Belgium and Aladangady et al. (2017) for the United States. This strong positive correlation is expected. Housing expenditures are a major component of total consumption, so if dwellings are a reasonable proxy for households (our preferred interpretation),⁹ then inequality in the distribution of housing values is naturally correlated with inequality in the income distribution. If preferences were homothetic, the correlation would be one. Given that households at the top spend relatively less on housing (Couture et al. 2024, Gaubert and Robert-Nicoud 2025), the correlation will not be one, but it should be high enough to infer meaningful information about income. Future research should more formally pin down the relationship between the two.

In addition to the connections with income inequality, the value of a dwelling is significantly influenced by local amenities and public goods (Baum-Snow and Hartley 2020, Couture et al. 2024). As highlighted by Figure A4, the price per square meter in Catalonia is substantially higher in Barcelona and its adjacent municipalities than in the rest of the region. It is also relatively high in coastal towns and the Val d’Aran, home to a renowned ski resort. These are, of course, no coincidences. At the core, housing prices reflect the value of a place (land, amenities, and local public goods). Housing prices impact people’s sorting, and through this sorting, they may influence the dynamics of prices themselves (Diamond and Gaubert 2022, Koster et al. 2016). In other words, housing prices are a natural and straightforward way to proxy for the value of land or, more generally, location (amenities, public goods, labor markets).

Relatedly, housing space inequalities are informative about real consumption

⁹There are limitations to this proxy exercise: mainly vacant houses (that contain no households) and homelessness (not observed in the data).

inequalities. As [Moretti \(2013\)](#) and [Diamond and Moretti \(2021\)](#) point out, after the cost of living (i.e., housing) is taken into account, income differences across US cities are not as stark as they seem at first glance. This important observation highlights the role of housing space as a measure of real consumption inequalities. To illustrate this point, note that one could decompose the value of a house into a location consumption component (land) and an actual housing consumption (structure) component. Housing space is related to the latter component, providing information about the real consumption differences between places.

The results in this paper are not informative about wealth inequality. Except at the very top, housing is the main asset for most households across the income and wealth distributions, making housing key to describing and understanding the dynamics of wealth inequality ([Kuhn et al. 2020](#), [Martínez-Toledano 2023](#), [Zucman 2019](#)). This is true in most Western countries and also in Spain, where capital gains, especially housing, have been the primary determinants driving growth in national wealth since the 1950s ([Blanco et al. 2021](#)). However, in a world where owner-occupiers and renters coexist, we cannot discuss wealth inequality without possessing ownership information, unfortunately not available in the public version of the data we could access for this paper.

5 Conclusions

This paper provided novel contemporaneous and long-run housing inequality estimates for Spain. Using the administrative data on the universe of dwellings in Spain and real estate transactions in Catalonia, we built measures of housing space and value inequality at all levels of aggregation and over time. Housing inequality in Spain today lies somewhere between Belgium and the United States, with the top 1% of the most spacious and valuable dwellings accruing over 6% of total dwelling space and value. Housing space inequality was much higher at the beginning of the 20th Century and experienced a significant

reduction after 1960, coinciding with a construction boom that more than doubled the housing stock in less than two decades. Housing value inequality in Catalonia increased between 2009 and 2019. First, due to a decrease in dwelling values at the bottom, likely stemming from a recession; then, due to increased values at the top, resulting from increased demand and a stagnant housing supply. Finally, we documented significant heterogeneity in housing space and value distributions across geographies and levels of aggregation, as well as a positive correlation between housing value and income inequality, suggesting a connection between the two.

The geographic heterogeneity we documented highlights how more readily available aggregated inequality measures may mask important distributional features of the underlying data. Therefore, this research, and especially the inequality estimates it introduces and makes publicly available through the WHID project, provides a new tool for researchers and policymakers to study the aggregate or local impacts of policies on housing-related outcomes. Alternatively, WHID estimates could be employed to proxy for local public goods or place characteristics.

This paper highlights at least two avenues for future work. First, for Spain, in particular, obtaining restricted ownership information from the cadastre and nationwide transaction data would allow for an expansion of value estimates and a discussion of wealth. Finally, future research should more formally establish a connection between housing value, housing space, and housing wealth inequality, as well as income and wealth inequality.

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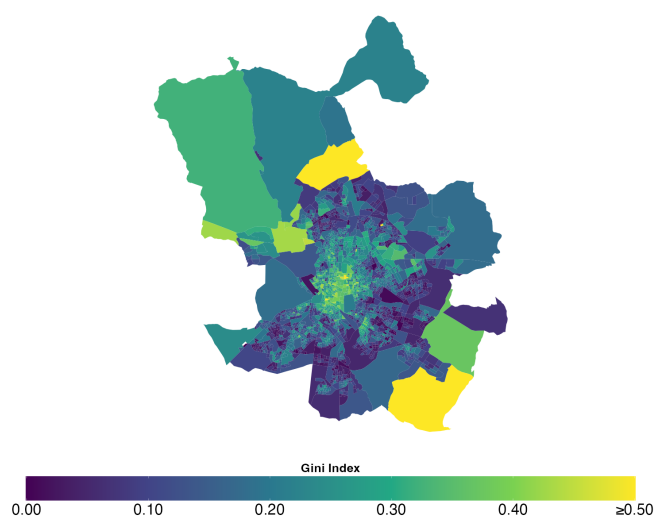
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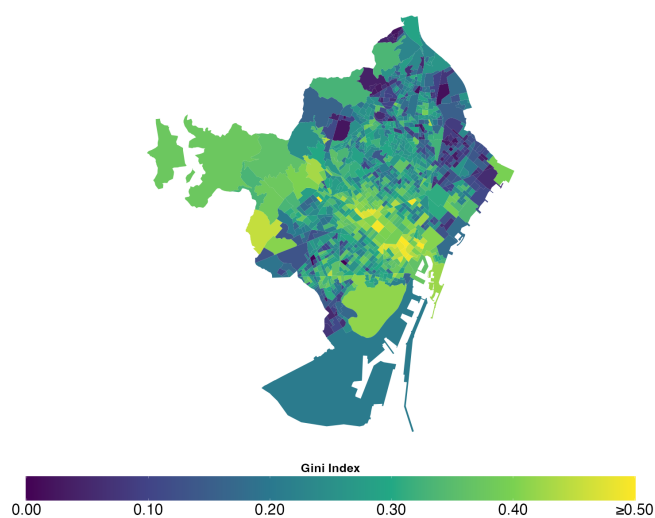
Web Appendix of Housing Inequality in Spain

Gerard Domènech-Arumí and Giovanni Paolo Mariani

A Supporting Figures and Tables



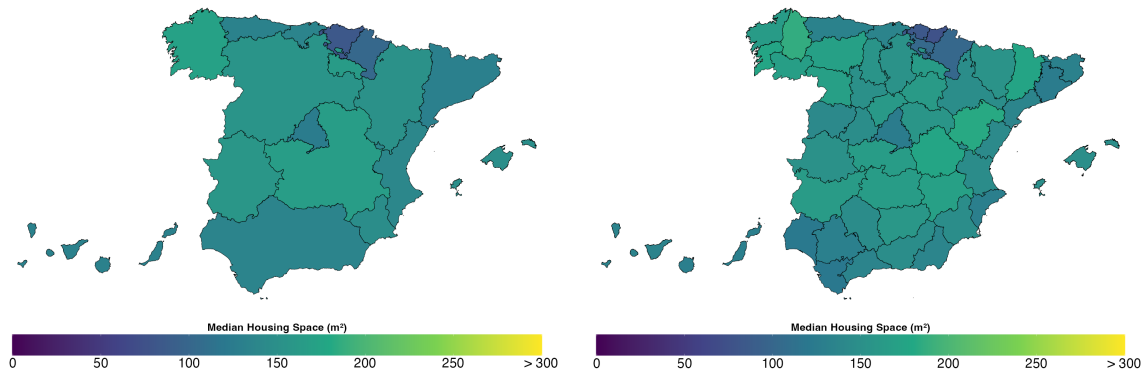
(a) Madrid



(b) Barcelona

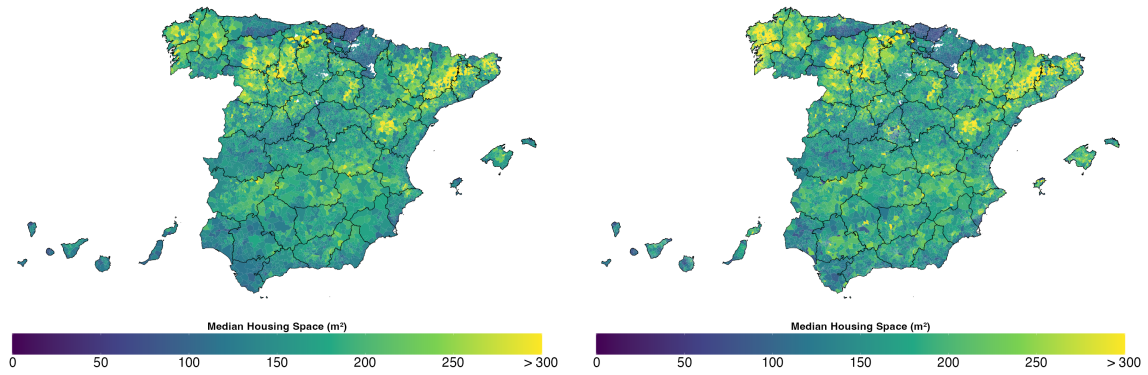
Figure A1: Housing space Gini in Madrid and Barcelona (census tract level)

Notes: Housing space is measured in square meters.



(a) Region

(b) Province



(c) Municipality

(d) Census tract

Figure A2: Median dwelling space (m²) in Spain

Notes: Each map illustrates the median dwelling size (m²) at a given subnational level in 2023. Darker colors describe geographic units with smaller dwellings. Brighter colors denote areas with larger dwellings. The median dwelling space in Spain is 137m² (2023).

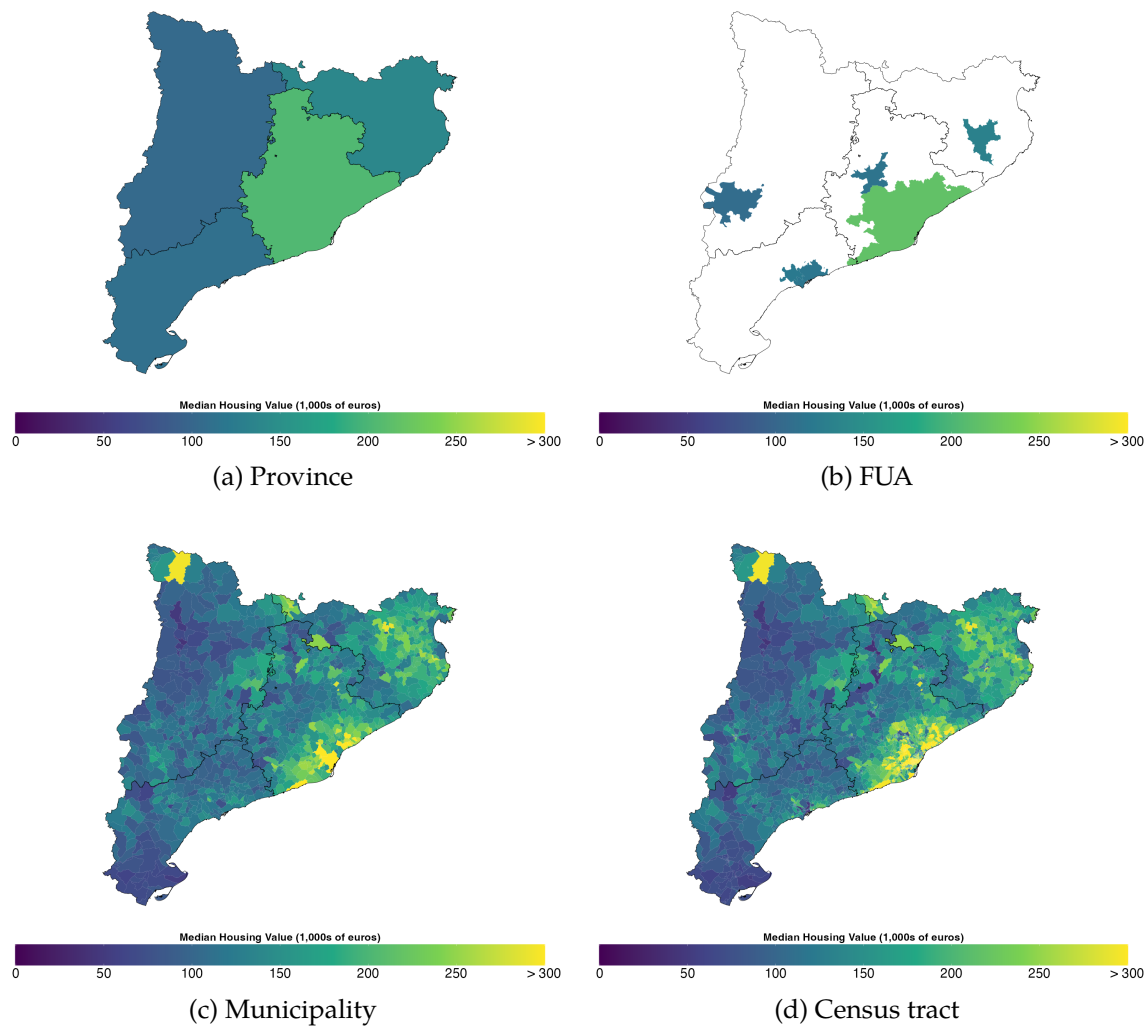


Figure A3: Median housing value in Catalonia

Notes: Each map illustrates the median housing value (1000s of euros) at a given subregional level in 2019. Darker colors describe geographic units with lower dwelling values. Brighter colors denote higher values. The median housing value in Catalonia is 169,400 euros (2019).

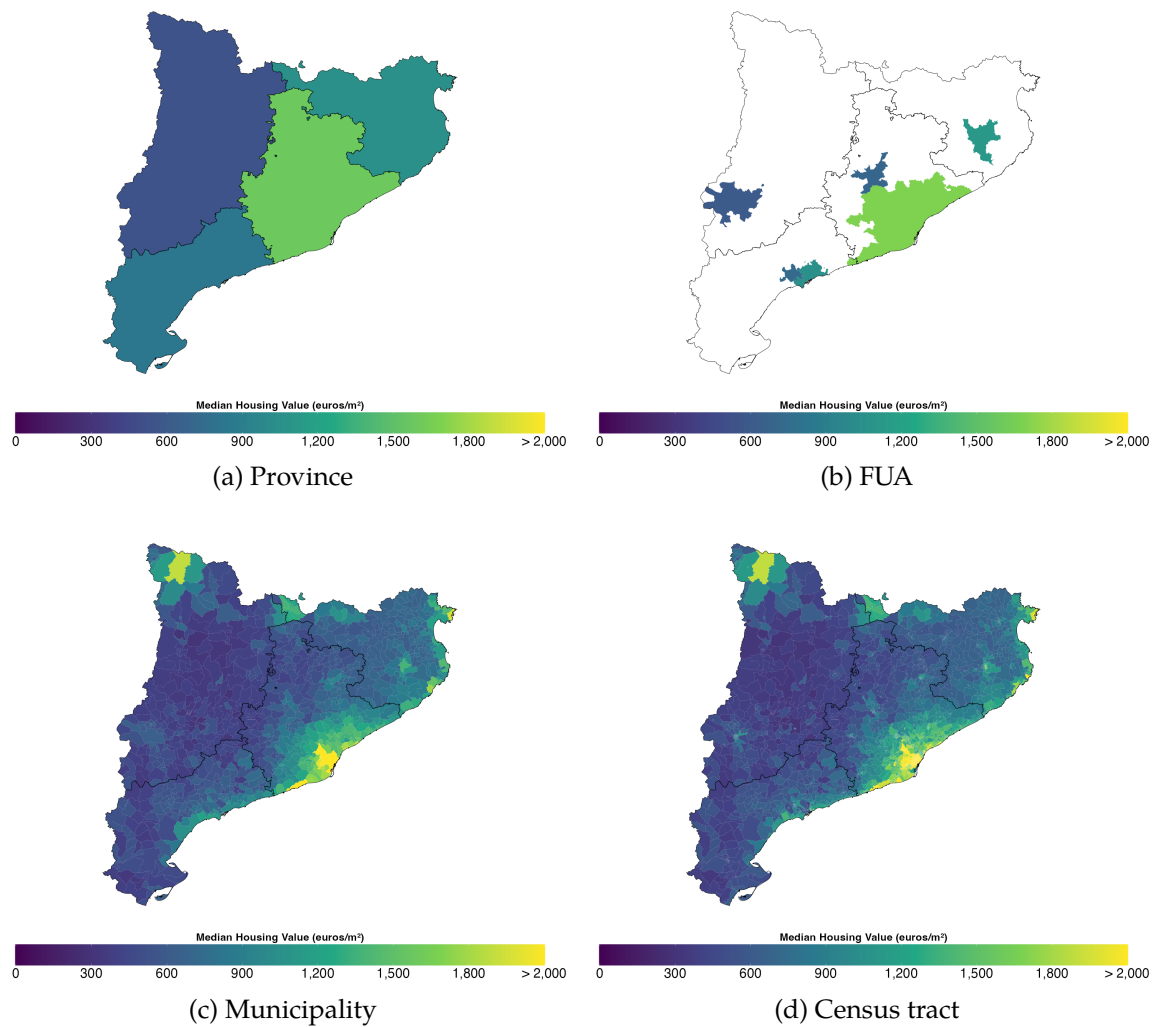


Figure A4: Median housing price per square meter in Catalonia

Notes: Each map illustrates the median housing value (euros per square meter) at a given subregional level in 2019. Darker colors describe geographic units with lower values. Brighter colors denote higher values. The median price per square meter in Catalonia is 1,295 euros (2019).

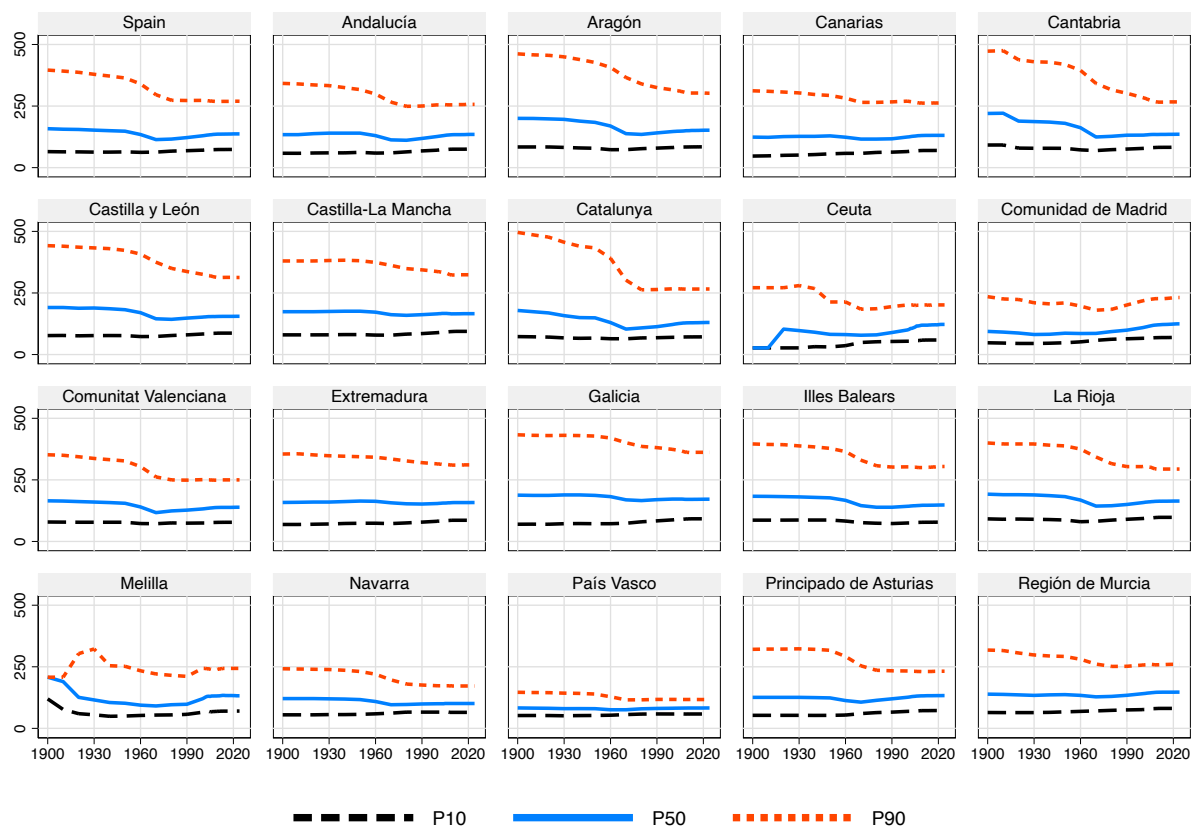
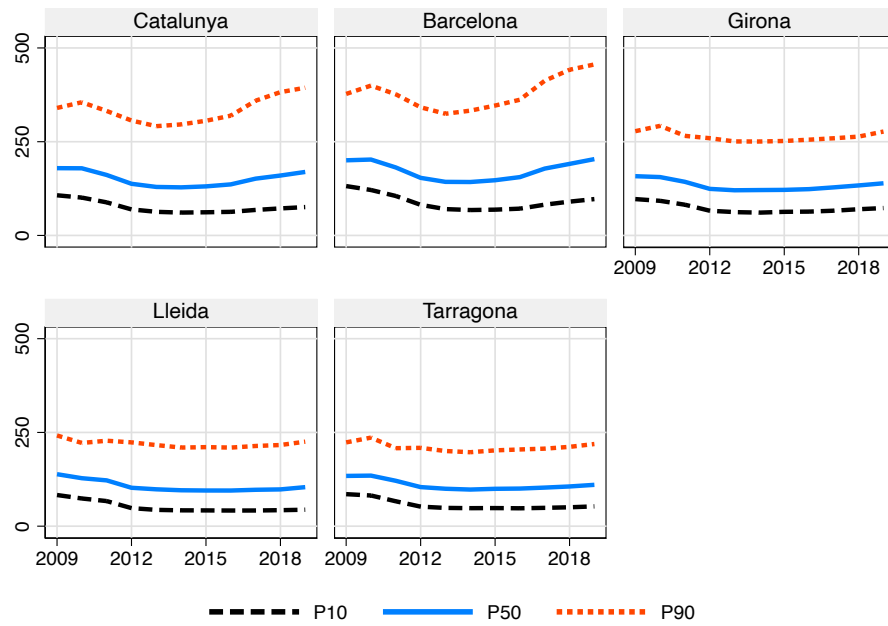
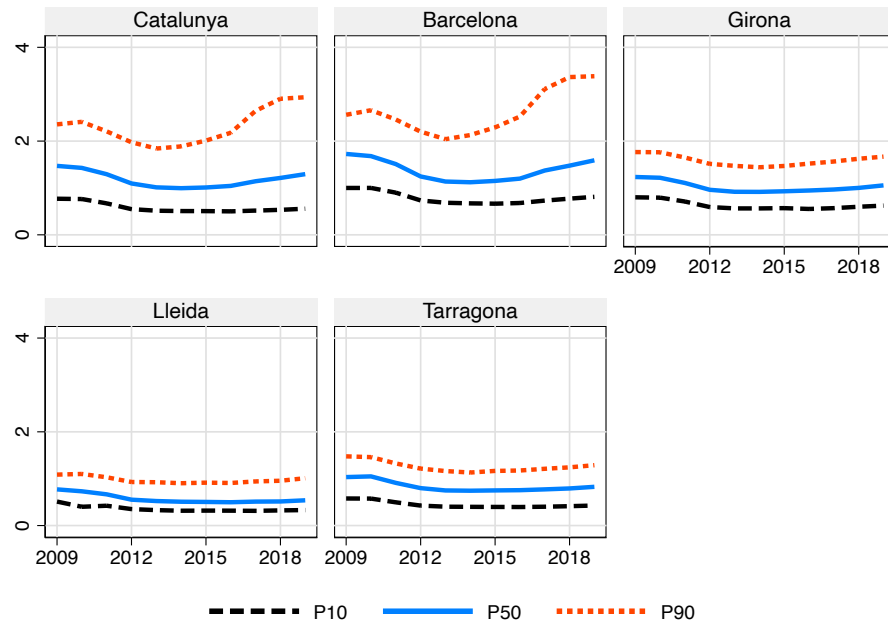


Figure A5: Evolution of housing space inequality (P10, P50, and P90) across regions

Notes: Housing space is measured in square meters. The top-left panel describes the evolution of housing space inequality (P10, P50, and P90) for Spain. The rest of the panels report the evolution of housing space inequality at the regional level.



A. Housing value



B. Price per square meter

Figure A6: Evolution of housing value inequality (P10, P50, and P90) in Catalonia

Notes: Housing value and price per square meter are measured in 1000s of euros. The top-left panel describes the evolution of housing value inequality (P10, P50, and P90) for Catalonia. The rest of the panels report the evolution of housing space inequality at the province level.

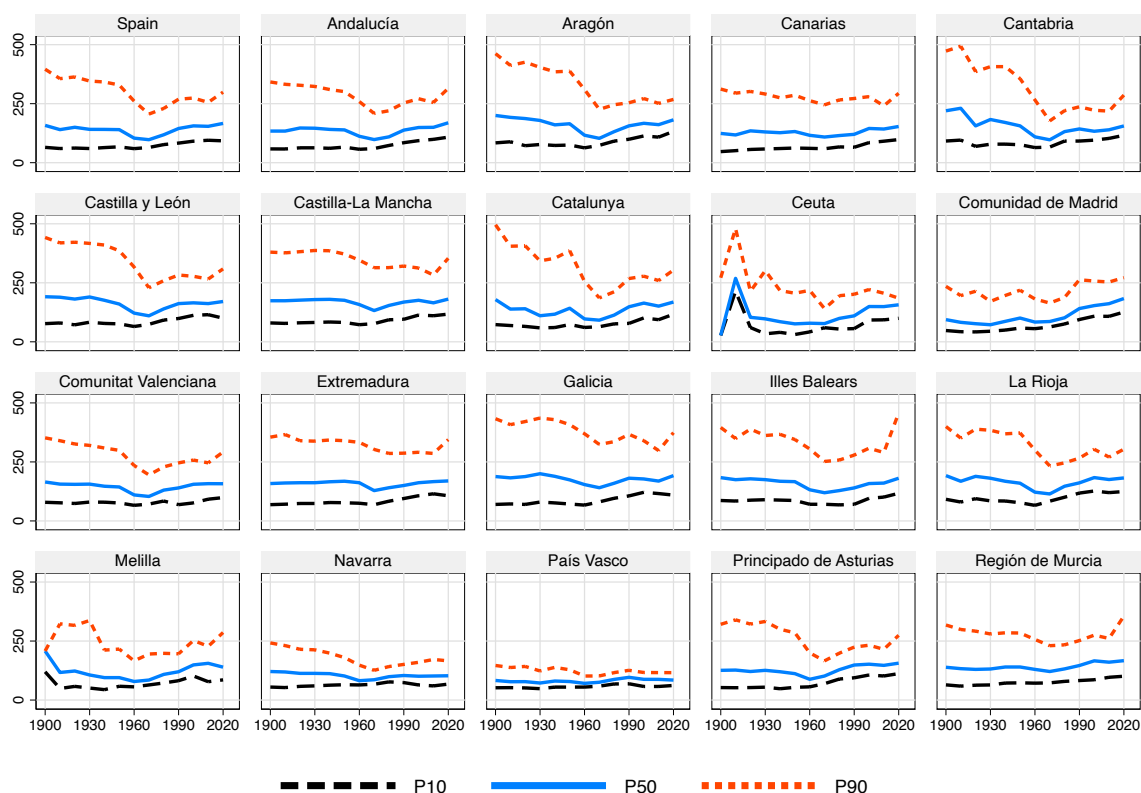


Figure A7: Housing space inequality (P10, P50, and P90) by decade-cohort across regions

Notes: Housing space is measured in square meters. The top-left panel describes the evolution of housing space inequality (P10, P50, and P90) by decade-cohort for Spain. The rest of the panels capture the evolution of the same outcome for the regions. Estimates are constructed based on the dwellings entering the housing stock in the previous decade (e.g., the observation for 1910 is based on the dwellings constructed between 1901 and 1910).

Table A1: Housing inequality in Spain: Theil decomposition

	Theil	Theil-Between	Theil-Within
A. Housing space			
A.1 Country			
Spain	0.184	0.014	0.17
A.2 Commuting Zone (Top 12)			
Alicante	0.122	0.034	0.088
Barcelona	0.204	0.048	0.156
Bilbao	0.07	0.034	0.036
Córdoba	0.186	0.077	0.109
Las Palmas	0.169	0.042	0.127
Madrid	0.165	0.075	0.09
Murcia	0.121	0.023	0.098
Málaga	0.15	0.042	0.109
Palma de Mallorca	0.183	0.038	0.146
Sevilla	0.159	0.05	0.109
Valencia	0.112	0.029	0.083
Zaragoza	0.135	0.045	0.09
B. Housing value			
B.1 Region			
Catalunya	0.259	0.035	0.225
B.2 Commuting Zone			
Barcelona	0.227	0.134	0.092
Girona	0.186	0.052	0.134
Lleida	0.214	0.042	0.172
Manresa	0.175	0.029	0.146
Reus	0.209	0.051	0.158
Tarragona	0.179	0.067	0.113

Notes: *Theil* is the Theil index. *Theil-Between* and *Theil-Within* decompose the Theil index into a between and within component. In Panels A.1 and B.1 (Inequality measured at the country and region level), the Theil index is decomposed to capture existing inequality between and within provinces. In Panels A.2 and B.2 (commuting zone level), the Theil index is decomposed to capture existing inequality between and within statistical census tracts. Housing space is measured in square meters. Housing value is measured in 1000s of euros (2019 estimates). *Commuting Zones* proxied by Functional Urban Areas (FUAs) (only the largest cities are included in this table).

Table A2: Housing value (price per square meter) inequality in Catalonia (2019)

	Gini	Top 1	Top 10	Bottom 50	P99	P90	P50	P10
A. Region								
Catalunya	0.321	0.029	0.237	0.278	4117	2935	1295	558
B. Province								
Barcelona	0.281	0.024	0.208	0.304	4233	3383	1590	811
Girona	0.207	0.023	0.174	0.351	2303	1669	1055	624
Lleida	0.251	0.04	0.214	0.328	1893	1010	539	331
Tarragona	0.226	0.021	0.173	0.333	1667	1288	827	427
C. Commuting Zone								
Barcelona	0.251	0.023	0.196	0.324	4265	3457	1690	1027
Girona	0.197	0.018	0.16	0.354	1998	1765	1125	660
Lleida	0.212	0.021	0.174	0.349	1229	1028	599	368
Manresa	0.177	0.018	0.153	0.37	1197	1021	691	427
Reus	0.183	0.018	0.162	0.367	1371	1179	735	489
Tarragona	0.178	0.018	0.155	0.375	1777	1442	1050	538

Notes: *Gini* is the Gini index. *Top 1* and *Top 10* measure the share of housing value that accrues to the top 1 or 10% most valuable dwellings in the specific geography, respectively. *Bottom 50* measure the share of housing value accrued to the 50% least valuable dwellings. *P99*, *P90*, *P50*, and *P10* indicate the value of the dwelling in the 99th, 90th, 50th, and 10th of the dwelling value distribution in the specific geography, respectively. Housing value is proxied by dwelling price per square meter (in euros). *Commuting Zones* proxied by Functional Urban Areas (FUAs).

B Data Appendix

B.1 Cadastre

B.1.1 Data description

The cadastre (*catastro*) is an administrative registry on Spanish real estate hosted and maintained by the Ministry of the Treasury (*Ministerio de Hacienda*) for taxation purposes. The data provides a snapshot of all real estate at a point in time, late 2023 in our data.

The central cadaster registry (*Dirección General del Catastro*) contains the data for the 17 regions in the common fiscal regime (*régimen fiscal común*), which includes all (15) regions and (2) autonomous cities, except the Basque Country and Navarra. These two regions are under a special tax regime (*régimen fiscal especial*), which allows their government bodies to administer their taxes and host their separate cadastral registries.

We obtained and harmonized cadastral data from the three cadaster registries, thus covering the whole territory.

B.1.2 Data cleaning

Common fiscal regime regions. For the 15 regions and two autonomous cities in the common fiscal regime, we utilize data made accessible via the Catastro INSPIRE framework,¹ which simplifies data and provides detailed information on cadastral parcels, buildings, and addresses.

We restrict our attention to residential parcels with at least one dwelling. For each building with at least one dwelling, we observe the year of construction, geographic coordinates (latitude and longitude), total residential space, land area, number of stories, number of dwellings, and construction quality.

The data does not contain specific information on individual dwellings within a building. Therefore, we have to impute a “dwelling space,” which we do by assuming an equal distribution of space across all dwellings within a building.

$$\text{Dwelling Space} = \frac{\text{Residential Space in the Building}}{\text{Number of Dwellings in the Building}} \quad (3)$$

Navarra. Cadastral data in Navarra are partially available through INSPIRE. However, the INSPIRE repository appears to misreport the residential area information. In practice, this means that it is not possible to retrieve all the necessary information from a single

¹The INSPIRE Directive aims to establish a European Union spatial data infrastructure, facilitating the sharing of environmental spatial information among public-sector organizations, enhancing public access to spatial data across Europe, and supporting policy-making across administrative boundaries.

centralized location for that region. We had to supplement that data with municipal-level cadastral data (with dwelling space information) provided by the regional government on their website.²

Unlike the regions in the common fiscal regime, the Navarra data is more precise as it provides information at the dwelling level. Specifically, for Navarra, we observe the exact size of all dwellings. This implies that dwelling space imputations are, in principle, not needed, which would allow us to study housing space distributions in more detail. However, exploiting this richer feature in the Navarra data would make our estimates for Navarra not comparable to those for the rest of Spain. Therefore, we opted not to use the detailed dwelling-level information for harmonization purposes. Instead, we impute sizes by assigning each dwelling the average dwelling space in the building, as for the rest of Spain.

Basque Country. The Basque Country maintains three separate cadastral registries corresponding to its three provinces: Bizkaia, Álava, and Gipuzkoa. Similar to the Navarra registry, we retrieve municipal-level data for each province, which also reports information at the dwelling level.³ For harmonization purposes, we apply the same aggregation procedure as in Navarra, summing residential areas at the building level and uniformly distributing the average residential area across dwellings within each property.

Merging, geolocation, and cleaning. After merging the distinct regional datasets from the three registries, we obtain a comprehensive national database consisting of approximately 25.5 million dwelling-level observations. This number is close but slightly (4%) below the dwelling count provided by Spain's National Statistical Office (*Instituto Nacional de Estadística*, INE), which estimated the country had a total of 26.6 million dwellings in its latest census (2021).

The discrepancy between INEs and our figures based on the cadastre is likely a consequence of occasional inconsistencies in how dwellings are identified within a building. One of the exercises we carried out to validate the cadastral data was randomly

²The data is available here: <https://catastro.navarra.es/descargas/>. Note: the website can only be accessed from Spain.

³The websites of the cadasters for the three official provinces are: <https://appsec.ebizkaia.eus/O4GC000C/vistas/descargaParcelario.xhtml?language=es> for the Bizkaia Cadaster, [urlhttps://catastroalava.tracasa.es/descargas/?lang=en](https://catastroalava.tracasa.es/descargas/?lang=en) for the Álava Cadaster and <https://www.gipuzkoairekia.eus/es/datu-irekien-katalogoa/-/openDataSearcher/search/true> for the Gipuzkoa Cadaster. Regarding the latter, the website provides various datasets, so users need to specifically search for the cadaster data, which are available at the municipality level.

cross-checking some units with Google Maps and other data sources. While the vast majority of dwellings we checked seemed “okay” (e.g., based on Google Street View), we sometimes identified apartment buildings with multiple dwellings that only contained a single dwelling in the cadastre. These occasional measurement errors likely led to some degree of dwelling undercounting. Unfortunately, we were unable to identify any systematic pattern in the data associated with this misreporting issue, and therefore, we were unable to address the problem.

Since we are interested in studying housing at different levels of aggregation, we use the latitude and longitude information available in the data to link each dwelling to the seven existing administrative levels in the country: census tract, census district, municipality, functional urban area, province, region, and country. We use shape files with 2020 boundaries.

We followed a few additional cleaning steps. We dropped a small number of properties that we could not precisely geolocate. Specifically, 17 properties from the Spanish cadaster (covering the common fiscal regime regions) were wrongly geolocated to Navarra (1) and Basque Country (16). This problem likely arises due to minor mismatches between parcel boundaries in the cadastral data and administrative boundaries as defined by INE. Given their negligible share (approximately 0.0002% of total parcels), we exclude these observations from our analysis as their correct census tract assignment could not be retrieved.

We also excluded properties that could not be geocoded, primarily due to inaccuracies in the recorded latitude and longitude data. This issue was present in 926 properties in the Spanish cadastral data (0.009%). Comparable geocoding errors were present in the other cadastral registries, with non-geocodable properties accounting for 0.02% in Navarra, 0.003% in Álava, 0.0007% in Bizkaia, and 0.001% in Gipuzkoa. We exclude these properties from our analysis.

Additionally, we dropped properties for which we suspect the residential space attribute is wrong — implausibly large or small. Specifically, we exclude dwellings with areas under 10 square meters or above 10,000 square meters. These properties we exclude represent approximately 0.02% of all observations in the final national dataset.

In terms of imputations, we have to impute a year of construction when it is missing (0.08% of dwellings). In these few instances, we use the median construction year in the census tract.

The final cleaned dataset contains 25,457,730 dwellings.

B.1.3 Discussion: accuracy of long-run estimates

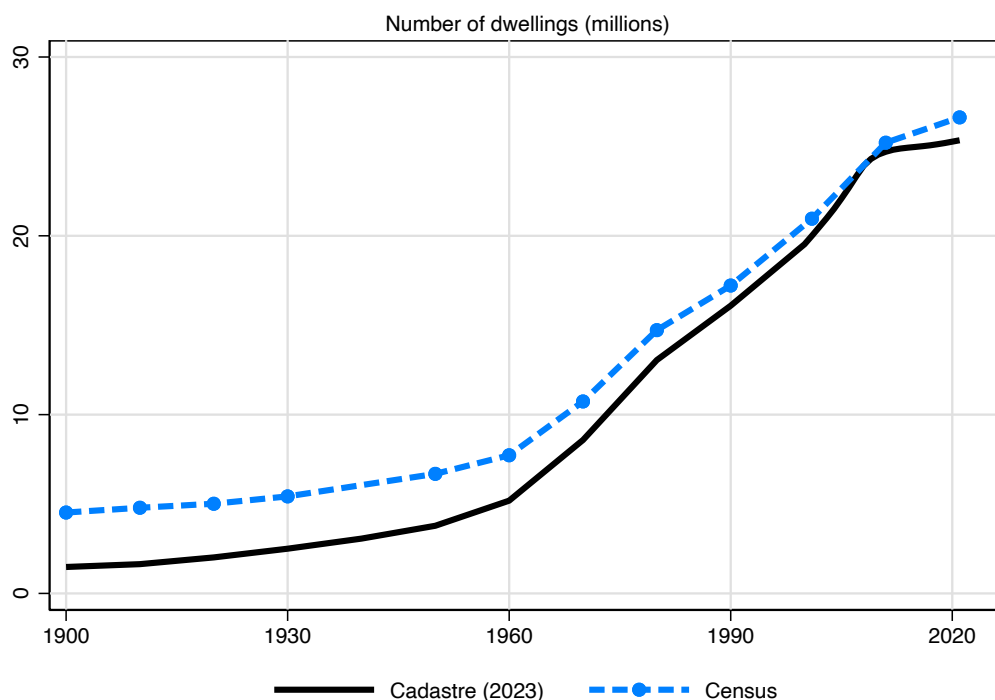


Figure B1: Evolution of the housing stock in Spain (1900-2021)

Notes: The black solid line describes the evolution of the housing stock based on a cross-section of cadastral data as of 2023. We use the year of construction to infer the dwellings present at a point in time. This series primarily misses demolished dwellings, which are not observed in the 2023 data. The dashed blue line shows the evolution of the dwelling stock based on official figures from the *Censo de Población y Viviendas* or *Censo de Edificios* (1950-2021). The figures for 1900, 1910, 1920, and 1930 are retrieved from the estimates in Carmona et al. (2014). We did not find an estimate for 1940.

Cross-sectional cadastral data can be used to analyze housing over time because (1) the year of construction is observed, and (2) housing (unlike people) does not move. The main caveat, repeatedly stressed throughout the paper, is that dwellings exiting the stock (e.g., demolitions) are not observed. This section discusses the reliability and possible bias of our long-run estimates.

Figure B1 illustrates the evolution of the housing stock in Spain, based on our cross-sectional cadastral data extracted in 2023 (black solid line) and census data (1950-2021), as well as estimates by Carmona et al. (2014) (1900-1930) (blue dashed line). We make four observations.

First, the two lines closely track each other. Both series exhibit anemic growth in the stock from 1900 to 1960, a significant growth from 1960 to 2010, and a slowdown

thereafter. The cross-sectional data from 2023 captures the trends in the stock remarkably well throughout the whole period.

Second, the gap between the two lines is narrow in the most recent years and wider in the earlier ones. That was expected. A dwelling constructed in 1900 is more likely to have been demolished by 2023 than another constructed in 2020. In other words, the accuracy of our estimates is worse the farther back in time we go.

Third, relatedly, we capture a significant fraction of the housing stock, dating back as far as 1900. According to our data, 1.48 million dwellings were built in 1900 or before. There is no official data on the Spanish housing stock before 1950, but Carmona et al. (2014) estimate there were approximately 4.52 million dwellings in 1900. This implies that our approach captures roughly a third of the dwellings that existed at the time. We would capture over two-thirds of the stock after 1960 and over 90% post-1990.

Fourth, the gap between the two lines never closes, even in the most recent period. This is due to two main factors. On the one hand, different estimates come from different sources. The cadaster is based on administrative records from the Ministry of the Treasury. The census estimate is based on surveying a sample of the population and buildings, with varying data collection methodologies employed throughout the period, including a recent change in the 2021 census. The cadastre data should, in general, be more accurate (for the extraction year, 2023). On the other hand, we identified some degree of measurement error in the cadastral data (see Section B.1.2), and we dropped a few dwellings (6,461) during the data cleaning process when their size was below 10 square meters, as we assumed the record for those dwellings was probably wrong.

Based on the previous observations, we make the following remarks:

First, the primary concern in the early estimates is selection. Even if our data only contains a subset of all the dwellings that existed at a point in time (e.g., 33% in 1900), we can still make inferences about population objects (e.g., housing space Gini in 1900). However, sample selection is likely to be a major concern in this inference exercise. We are certain that the subsample of dwellings built before 1900 that survived until 2023 is not a random sample of the stock. At the very least, they are likely to be positively selected in terms of quality – they survived, so they probably had a better structure than those that did not.

Second, our long-run estimates are likely to underestimate housing inequality. Due to a lack of reliable data, it is impossible for us to tell the direction of the bias. Still, if we believe the dwellings that did not survive were predominantly in the lower tail of the distribution (in space or value), then our long-run estimates should be underestimating inequality.

Third, we are significantly more confident in the quality of the estimates post-1940 for two reasons. On the one hand, we're capturing over 50% of the stock post-1940, which alleviates selection problems. On the other hand, the close alignment in the trends displayed in Figure B1 suggests that we are capturing most of the dwellings entering the stock in the second half of the 20th Century.

Fourth, we are significantly less confident in the pre-1940 due to three factors. First, based on available estimates, we are capturing less than 50% of the stock that existed at the time with our cadastral data, which exacerbates selection concerns. Relatedly, sample representativity and selection are likely to vary across locations, which makes the interpretation of estimates below the national level even trickier. For instance, we strongly suspect that our early estimates for Ceuta and Melilla are especially biased based on the discrepancy in the number of dwellings we observe in the cadastre and the number of people that lived in the city at the time (e.g., we estimate Ceuta had 64 dwellings in 1900, but it is estimated that approximately 13,000 people lived in the city in that year). Finally, an undetermined number of dwellings localized in specific areas were destroyed during the Spanish Civil War (1936-39).

Finally, despite the previous concerns, we see this as a valuable exercise. First, because, with the necessary caveats, there is something we can learn from it. Second, future research might be able to correct some of the biases present in the estimates.

B.2 Agència Tributaria de Catalunya

The Agència Tributaria de Catalunya (ATC) data contains information on the universe of real estate transactions in Catalonia from January 2009 to December 2019. This data is generated from a levy on property transfers (*Impuesto de Transmisiones Patrimoniales*, ITP). In total, there were approximately 275,000 transactions over the sample period, with 65,000 taking place in Barcelona.

For each transaction, the data contains information on the sale price, dwelling characteristics (e.g., space), and location. Critically, all properties in this dataset can be easily linked to the cadastre using a unique identifier.

This data is confidential and can only be accessed by requesting the data directly from the ATC.

B.3 Other data

This paper uses several additional datasets. These include:

- **INE Municipal Registry (2004-2022).** This data contains population counts broken down by sex and nationality at the census tract level.
- **INE Historical Population Counts (1900-2001).** Decennial population counts, available at the province level.
- **INE Atlas de la Renta (2016-22).** This dataset contains a number of income distribution statistics (e.g., median after-tax income), available at the census tract level and above.
- **2011 Census.** This dataset contains detailed socioeconomic and dwelling information at the census tract level for 2011. We only use this data to study the correlations between different sociodemographic characteristics of places and housing inequality.
- **2011 National Election Results.** This dataset contains the electoral results for the 2011 National Election, with information available at the census tract level, retrieved from the Spanish Ministry of Interior website. We only use this data to study the correlations between different sociodemographic characteristics of places and housing inequality.

All the datasets listed above are publicly available on the INE website.

B.4 Housing value imputations

We employ transaction-level data from the ATC, comprising 275,833 sales across Catalonia from 2009 to 2019, to train a predictive model for estimating dwelling value. Specifically, we used the Ranger package in R to implement Breiman (2001)’s random forest algorithm to predict the log price per square meter. The literature has shown that random forests perform especially well in predicting housing prices (Mullainathan and Spiess 2017). When training our model, we use 14 main dwelling-related predictors, including nonlinear polynomial transformations of key housing characteristics (e.g., building age, size). To precisely capture location effects, we also include a census tract factor variable and variables describing the latitude and longitude of a dwelling. To account for temporal and regional variation flexibly, our specification also incorporates year fixed effects interacted with province indicators. Table B1 describes the main variables used in the prediction.

To enhance the predictive performance of the Random Forest model, we performed hyperparameter tuning with a focus on three core parameters. First, we adjusted the

Table B1: Variables used in the Random Forest model

Variable	Description
CUSEC	Census tract identifier
CUMUN	Municipality identifier
CPRO	Province identifier
CPRO_year	Province-year interaction term
Lat	Latitude coordinate
Lon	Longitude coordinate
age	Age of the property at transaction date
year_construction	Year the property was constructed
first_sale	Indicator for whether the property was sold for the first time
year_f	Year of the transaction
space_d	Surface area of the property (m ²)
space_d2	Surface area squared (m ⁴)
sqrt_space	Square root of surface area
log_size	Logarithm of surface are

number of predictors considered at each decision split, which controls model complexity and the number of variables selected. Second, we tuned the minimum number of observations required in each terminal node, which regulates tree depth and helps prevent overfitting. Third, we optimized the subsample proportion of the data used to grow each tree, a parameter that affects model variance and computational efficiency. The tuning process involved a grid search over plausible ranges for each parameter, with model performance assessed using the out-of-bag root mean squared error (OOB RMSE) as the selection criterion. Specifically, we tested three levels for the number of predictors per split, three levels for the minimum terminal node size, and two levels for the subsample proportion, yielding a total of 18 model combinations. The optimal configuration selected for the final model yields 8 predictors sampled at each split, larger minimum terminal node sizes (9), and a training subsampling of 75% of the available data for each tree.

Our model yields an out-of-bag root mean squared error (OOB-RMSE) of 0.12, which lies in the lower end of the spectrum in terms of common prediction errors reported in the literature, often ranging between 0.10 and 0.16 (Čeh et al. 2018, Fan et al. 2006, Mullainathan and Spiess 2017).

Figure B2 reports the in-sample fit of our predictions. The high correlation between the predicted and actual prices, as displayed in the figure, provides reassurance regarding our predictions. However, the figure also suggests that our model slightly overestimates prices at the lower end and underestimates those at the upper end of the distribution,

indicating that we may be underestimating housing value inequality.

In terms of variable importance, our model indicates that location-related predictors are the most relevant, with the census tract, latitude, and longitude being the top three predictors in the model. Building age, housing space, and transaction year also emerge as highly important, albeit at a significant distance from predictors describing location.

We then used the trained model to predict the value of all dwellings in Catalonia (about 3.7 million), independently of whether they were transacted in 2009-19 or not. We predict a value for each dwelling for each year from 2009 to 2019, the years covered in the ATC data.

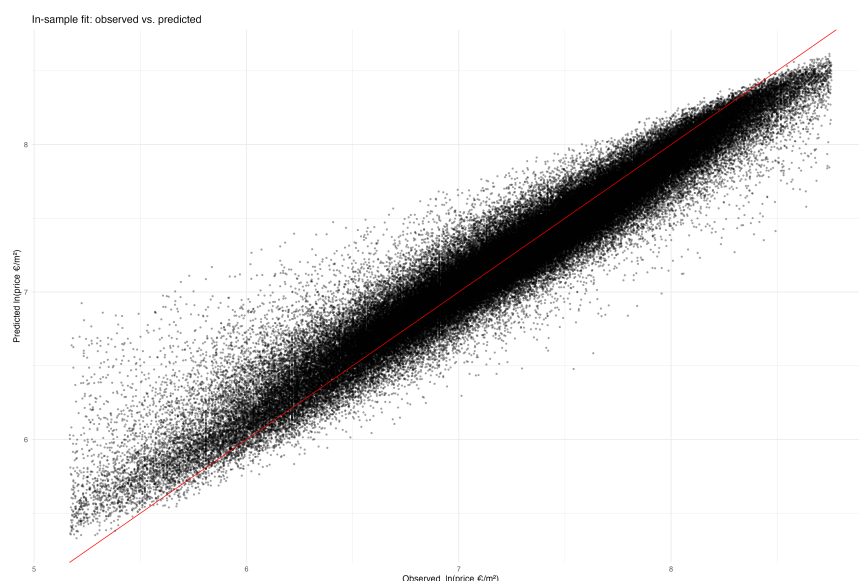


Figure B2: Prediction validity test: in-sample fit

Notes: Each dot is a real estate transaction. The x-axis reports the actual transaction price. The y-axis reports the model predicted price.

B.5 WHID – Spain

All housing inequality estimates are publicly available at doi.org/10.7910/DVN/HONBNB as part of the World Housing Inequality Database (WHID). The unit of observation in the data is a geography-year-dimension-type (e.g., region-2023-space-stock). Table B2 contains descriptions of all variables available in the dataset. Table B3 describes the time and geographic coverage of the data. All inequality estimates are available across all geographies. Value estimates are available yearly from 2009 to 2019 for the region of Catalonia only. “flow” estimates describe the characteristics of the dwellings entering the stock of housing in a specific decade (e.g.,

the row country-2020-space-flow for *gini* describe the Gini index of the space distribution of all dwellings that entered the housing stock between 2011 and 2020). Estimates are censored (set to missing) when the number of dwellings in a geography-year is 10 or less.

Table B2: WHID - Spain: variables description

Variable name	Variable description
year	Calendar year of the estimate
agglevel	Aggregation level of the estimate
agg	Aggregation level of the estimate
ineqvar	Inequality dimension
var_type	Variable type
ctry_id	Country ID (ISO)
ctry_name	Country name
reg_id	Region ID (NUTS 2)
reg_name	Region official name (Spanish)
prov_id	Province ID (NUTS 3)
prov_name	Province official name (Spanish)
fua_id	Functional Urban Area ID (EC/OECD)
fua_name	Functional Urban Area name
mun_id	Municipality (<i>Municipio</i>) numerical ID (INE)
mun_name	Municipality official name
cudis_id	Census district ID
cudis_name	Census district name
tract_id	Census tract name (INE)
gini	Gini index
share_top_01	Share of total dwelling value/space contained in the top 0.1% of most valuable/spacious dwellings
share_top_1	Share of total dwelling value/space contained in the top 1% of most valuable/spacious dwellings
share_top_10	Share of total dwelling value/space contained in the top 10% of most valuable/spacious dwellings
share_top_20	Share of total dwelling value/space contained in the top 20% of most valuable/spacious dwellings
share_bottom_50	Share of total dwelling value/space contained in the bottom 50% of least valuable/spacious dwellings
share_bottom_40	Share of total dwelling value/space contained in the bottom 40% of least valuable/spacious dwellings
share_bottom_20	Share of total dwelling value/space contained in the bottom 20% of least valuable/spacious dwellings
share_bottom_10	Share of total dwelling value/space contained in the bottom 10% of least valuable/spacious dwellings
theil	Theil index
theil_ctry_b_prov	Between-Theil index (variation between provinces) (available at country level only)
theil_ctry_w_prov	Within-Theil index (variation within provinces) (available at country level only; region for Catalonia – value estimates)
theil_ctry_b_tract	Between-Theil index (variation between census tracts) (available at FUA level only)
theil_ctry_w_tract	Within-Theil index (variation within census tracts) (available at FUA level only)
coef_var	Coefficient of variation
sd	Standard deviation
mean	Mean
p1	Value/Space of the dwelling in the 1st percentile
p5	Value/Space of the dwelling in the 5th percentile
p10	Value/Space of the dwelling in the 10th percentile
p25	Value/Space of the dwelling in the 25th percentile
p50	Value/Space of the dwelling in the 50th percentile (median)
p75	Value/Space of the dwelling in the 75th percentile
p90	Value/Space of the dwelling in the 90th percentile
p95	Value/Space of the dwelling in the 95th percentile
p99	Value/Space of the dwelling in the 99th percentile
p999	Value/Space of the dwelling in the 99.9th percentile
ratio9010	p90/p10
ratio9050	p90/p50
N_dwell	Number of dwellings in the geography/time period
censored	=1 if estimates are censored (less than 10 dwellings in the geography-year)

Table B3: WHID - Spain: geographic and time coverage

Variable name	Type	Values
year	Numeric	1900-2023
agglevel	String	ctry; reg; prov; fua; mun; cudis; tract
agg	Numeric	1 (country); 2 (region); 3 (province); 4 (FUA); 5 (municipality); 6 (census district); 7 (census tract)
ineqvar	String	space; value; value_sqm
var_type	String	stock; flow
ctry_id	Numeric	724
ctry_name	String	Spain
reg_id	String	01, ..., 19
reg_name	String	Andalucía, ..., Región de Murcia
prov_id	String	01, ..., 52
prov_name	String	Albacete, ..., Zaragoza
fua_id	String	Albacete, ..., Zaragoza
fua_name	String	Albacete, ..., Zaragoza
mun_id	String	02001, ..., 52001
mun_name	String	Ababuj, ..., Zújar
cudis_id	String	0200101, ..., 5200108
cudis_name	String	0200101, ..., 5200108
tract_id	String	0200101001, ..., 5200108015
tract_name	String	0200101001, ..., 5200108015

C Supplementary Results

This Appendix contains additional results not discussed in the main text, but that might be of interest to some readers.

Figure C1 shows the evolution of housing space inequality (Gini) across provinces. Figure C2 shows the evolution in the distribution of housing space (P10, P50, and P90).

Figure C3 shows the evolution of housing space inequality (Gini) by cohort across provinces. Figure C4 shows the evolution in the distribution of housing space (P10, P50, and P90) by cohort across provinces.

Figure C5 shows the evolution in the distribution of housing space (P10, P50, and P90) by cohort across the largest commuting zones (FUAs). Figure C6 plots the same information but includes all of Spain's commuting zones.

Table C1 describes housing space inequality across all of Spain's commuting zones (FUA) in 2023.

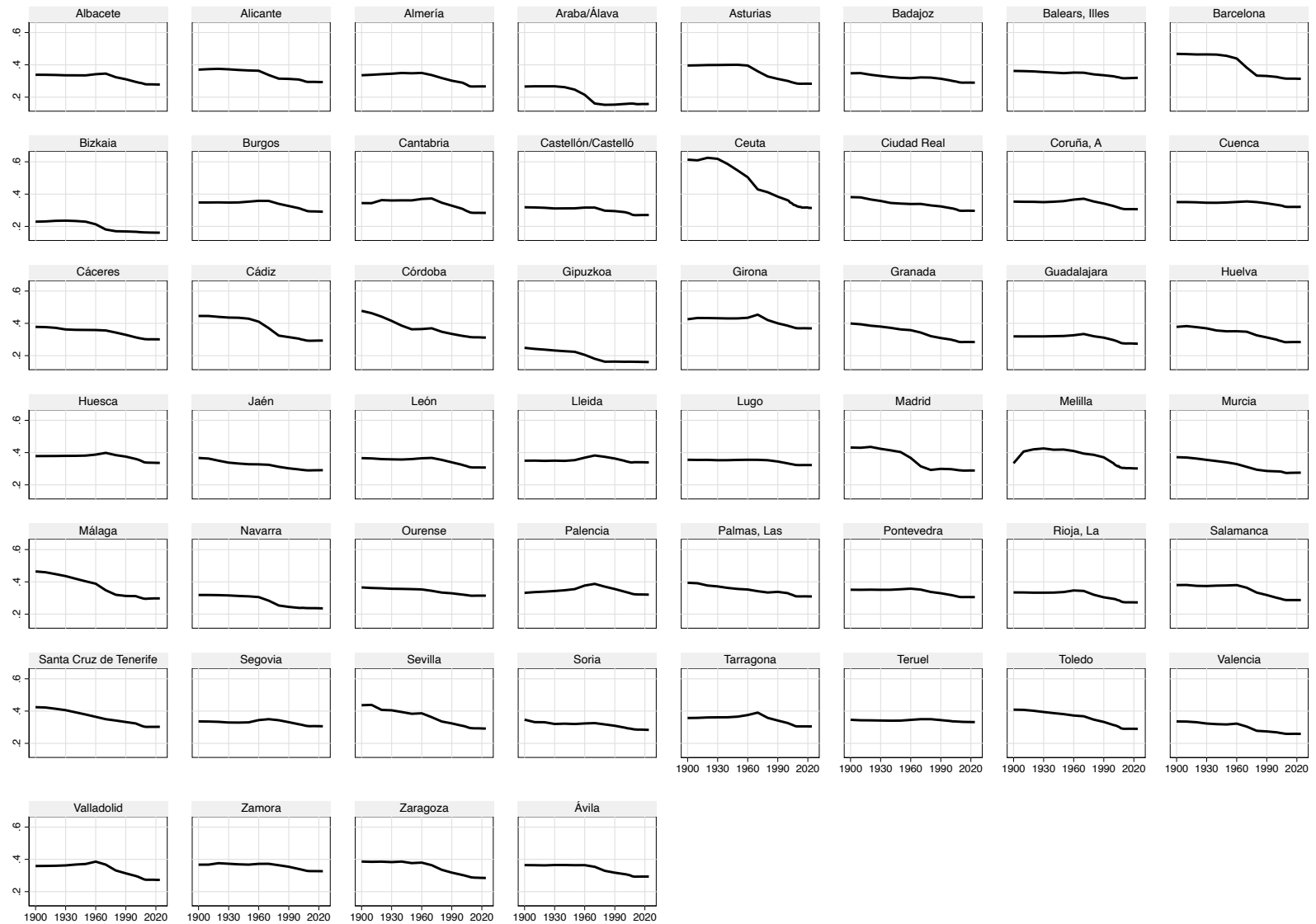


Figure C1: Evolution of housing space inequality across provinces

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (Gini) for a province.



Figure C2: Evolution of housing space inequality (P10, P50, P90) across provinces

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (P10, P50, and P90) for a province.

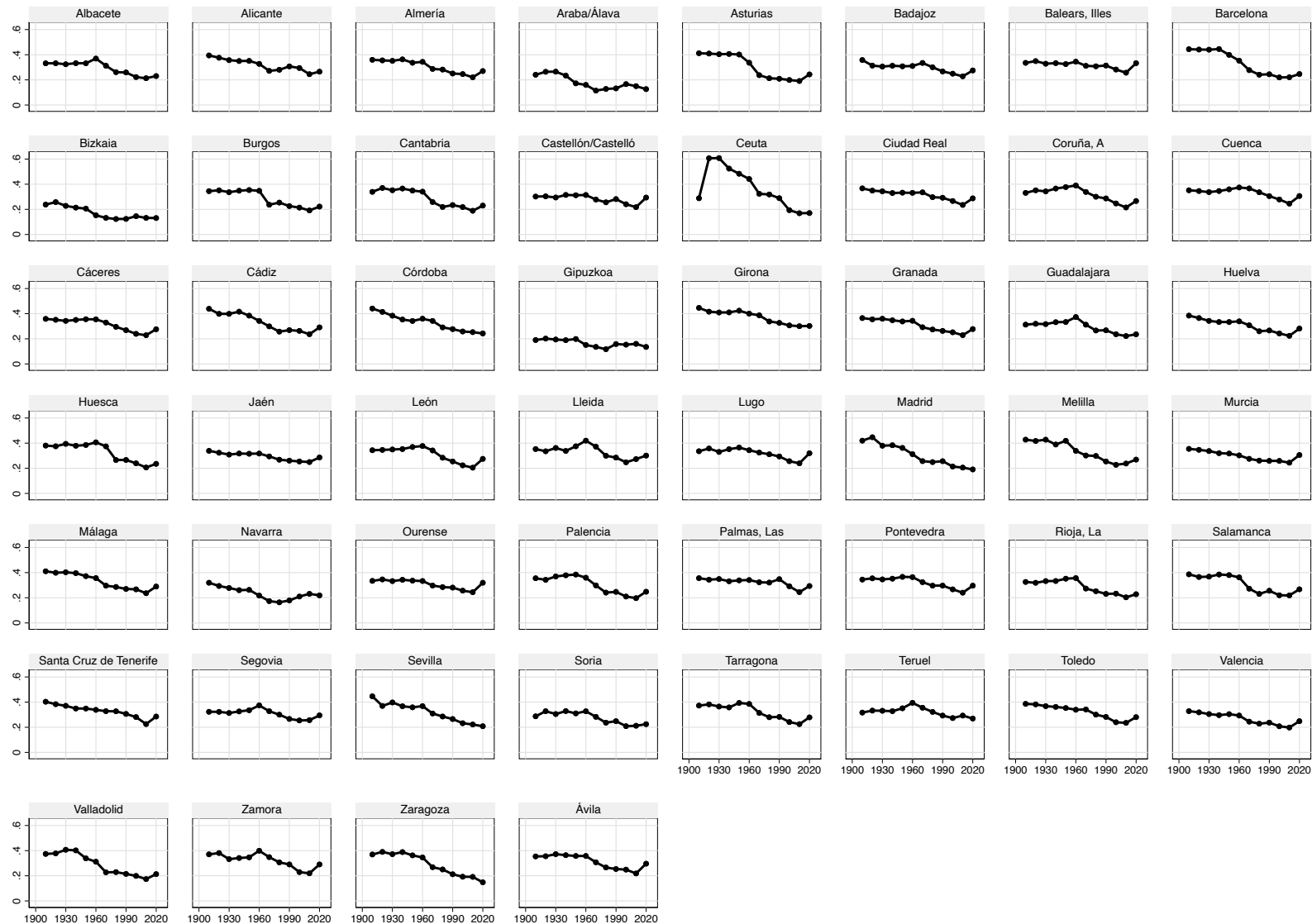


Figure C3: Housing space inequality (Gini) by decade-cohort across provinces

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (Gini) by decade-cohort for a province. Estimates are constructed based on the dwellings entering the housing stock in the previous decade (e.g., the observation for 1910 is based on the dwellings constructed between 1901 and 1910).



Figure C4: Housing space inequality (P10, P50, P90) by decade-cohort across provinces

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (P10, P50, and P90) by decade-cohort across Spanish provinces.

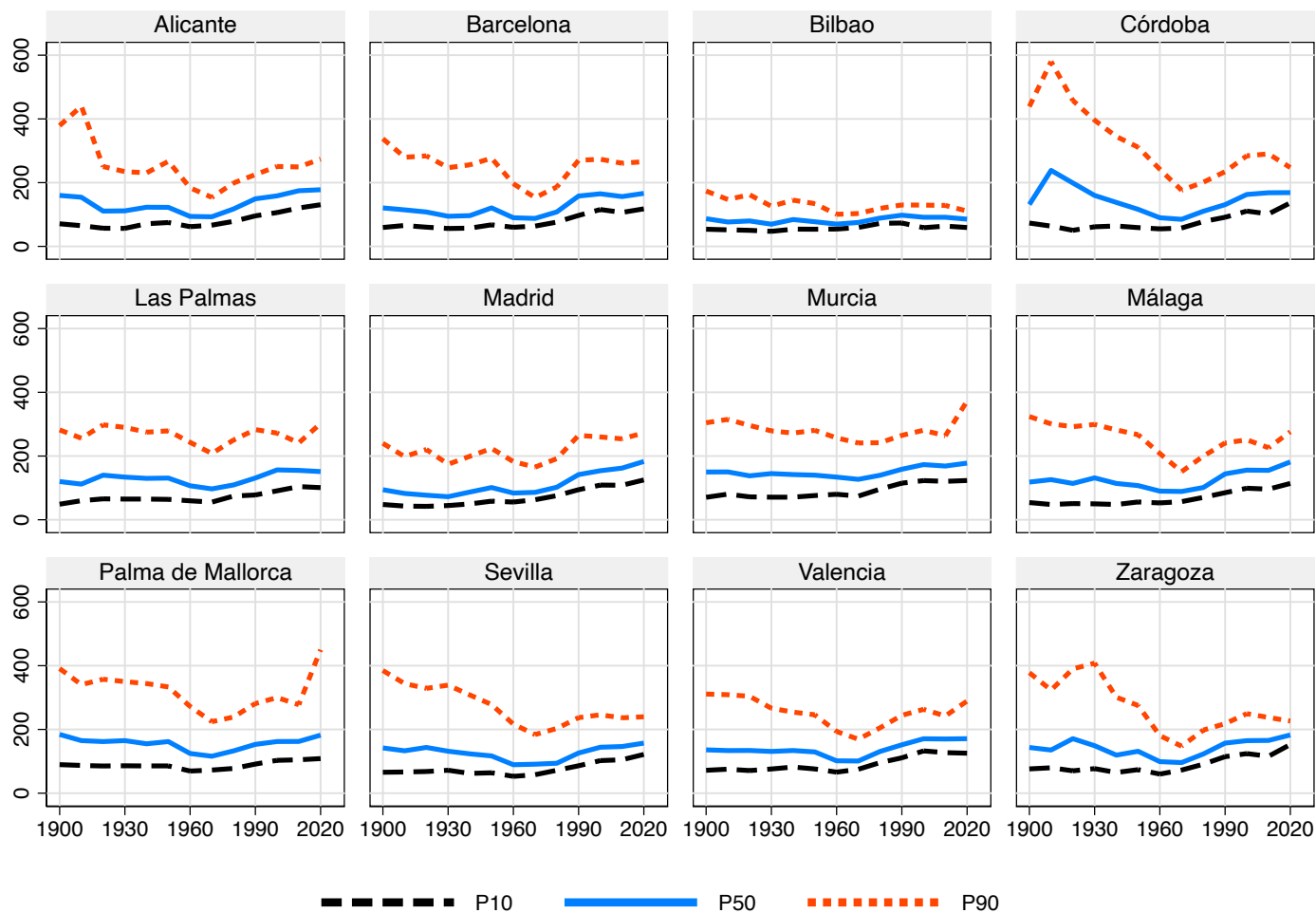


Figure C5: Housing space inequality (P10, P50, P90) by decade-cohort in top cities (FUAs)

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (P10, P50, and P90) by decade-cohort for the top Spanish commuting zones (FUAs). Estimates are constructed based on the dwellings entering the housing stock in the previous decade (e.g., the observation for 1910 is based on the dwellings constructed between 1901 and 1910).



Figure C6: Housing space inequality (P10, P50, P90) by decade-cohort across commuting zones (FUAs)

Notes: Housing space is measured in square meters. Each panel describes the evolution of housing space inequality (P10, P50, and P90) by decade-cohort across Spanish commuting zones (FUAs).

Table C1: Housing space inequality in Spain's Commuting Zones (FUA) (2023)

	Gini	Top 1	Top 10	Bottom 50	P99	P90	P50	P10
Albacete	0.239	0.048	0.218	0.344	543	269	160	100
Alcoy	0.307	0.074	0.266	0.3	703	266	144	78
Algeciras	0.253	0.052	0.225	0.333	459	215	130	79
Alicante	0.252	0.046	0.214	0.328	437	227	138	78
Almería	0.242	0.04	0.204	0.335	390	223	134	75
Arrecife	0.318	0.051	0.25	0.284	554	281	133	70
Avilés	0.252	0.048	0.22	0.331	456	211	132	74
Badajoz	0.253	0.051	0.23	0.335	541	250	145	91
Barcelona	0.305	0.072	0.261	0.299	530	233	119	71
Benidorm	0.288	0.056	0.242	0.308	468	205	116	60
Bilbao	0.184	0.038	0.19	0.378	234	122	83	59
Burgos	0.244	0.043	0.212	0.335	436	248	147	86
Cartagena	0.266	0.047	0.227	0.322	448	232	127	75
Castellón de la Plana	0.242	0.044	0.214	0.337	456	238	140	86
Ciudad Real	0.243	0.054	0.225	0.344	543	260	152	96
Coruña, A	0.272	0.054	0.243	0.324	599	274	152	89
Cuenca	0.251	0.047	0.216	0.333	462	244	151	82
Cáceres	0.248	0.049	0.219	0.336	500	250	152	87
Cádiz	0.241	0.063	0.227	0.342	453	182	113	75
Córdoba	0.306	0.06	0.257	0.296	562	245	130	68
Donostia/San Sebastián	0.159	0.034	0.174	0.396	171	102	73	55
Elche/Elx	0.284	0.054	0.241	0.311	542	235	130	73
Ferrol	0.28	0.047	0.234	0.313	553	293	154	88
Gandia	0.263	0.046	0.225	0.321	452	234	128	78
Gijón	0.24	0.052	0.226	0.343	484	205	126	83
Girona	0.357	0.078	0.291	0.263	694	259	121	61
Granada	0.259	0.048	0.227	0.329	508	260	146	90
Guadalajara	0.236	0.04	0.208	0.343	466	259	158	91
Huelva	0.253	0.048	0.225	0.331	421	201	116	72
Irun	0.192	0.035	0.191	0.374	218	117	78	52
Jaén	0.269	0.048	0.233	0.323	525	268	148	83
Jerez de la Frontera	0.273	0.059	0.237	0.316	487	213	117	75
Las Palmas	0.301	0.052	0.245	0.298	508	253	130	70
León	0.251	0.052	0.228	0.337	578	272	157	100
Linares	0.272	0.049	0.234	0.318	519	263	138	87
Lleida	0.294	0.059	0.256	0.308	727	313	165	98
Logroño	0.241	0.056	0.221	0.343	544	248	158	98
Lorca	0.301	0.063	0.262	0.307	785	329	175	95
Lugo	0.271	0.051	0.238	0.325	729	332	184	106
Madrid	0.289	0.056	0.24	0.304	478	234	126	70
Manresa	0.312	0.071	0.275	0.3	852	310	158	91
Marbella	0.317	0.073	0.275	0.297	745	271	142	78
Murcia	0.243	0.048	0.219	0.34	528	263	158	98
Málaga	0.276	0.053	0.232	0.313	471	223	129	71
Mérida	0.276	0.048	0.235	0.318	581	297	160	92
Ourense	0.278	0.043	0.229	0.318	555	301	165	86
Oviedo	0.241	0.053	0.221	0.345	481	223	142	82
Palencia	0.23	0.044	0.208	0.35	414	226	142	88
Palma de Mallorca	0.303	0.06	0.257	0.303	668	291	149	84
Pamplona/Iruña	0.208	0.043	0.209	0.379	290	150	98	68
Ponferrada	0.263	0.044	0.226	0.327	622	329	178	104
Pontevedra	0.285	0.047	0.239	0.312	668	364	177	104
Reus	0.289	0.063	0.251	0.311	710	278	151	88
Sagunto	0.218	0.041	0.205	0.357	425	218	140	88
Salamanca	0.225	0.043	0.21	0.352	418	227	133	89
Santa Cruz de Tenerife	0.288	0.052	0.241	0.308	539	270	140	80
Santander	0.262	0.048	0.232	0.328	501	244	135	82
Santiago de Compostela	0.319	0.05	0.263	0.29	813	394	171	104
Sevilla	0.282	0.055	0.24	0.312	484	229	126	72
Talavera de la Reina	0.273	0.054	0.238	0.32	596	275	155	90
Tarragona	0.278	0.058	0.238	0.315	478	215	123	70
Toledo	0.286	0.055	0.246	0.313	673	302	161	93
Torre Vieja	0.239	0.044	0.214	0.339	285	145	84	55
Valencia	0.239	0.045	0.215	0.34	462	232	142	86
Valladolid	0.24	0.051	0.215	0.341	458	236	147	87
Vigo	0.297	0.051	0.245	0.305	669	348	169	96
Vitoria-Gasteiz	0.163	0.042	0.186	0.396	286	140	103	78
Zamora	0.243	0.054	0.222	0.342	534	242	151	91
Zaragoza	0.247	0.056	0.222	0.335	464	222	139	83
Ávila	0.225	0.047	0.207	0.354	438	240	153	91

Notes: Commuting Zones proxied by Functional Urban Areas (FUAs). Housing space is measured in square meters.